Technical mechanics

Mechanical drive design calculation

Interactive teaching guide · AutoMethod v0.4.11 · 2026

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The guide covers both load input methods, cylindrical, bevel, worm, planetary and combined stages, no flexible transmission, belt, timing belt and chain options, and component checks. Numerical substitutions are a teaching example calculated by native engdoc; they do not replace the assigned task.

All equations are evaluated in SI. Speed n is in s^{-1}; $n_{min^{-1}} = 60 n_{s^{-1}}$. Stress is given in Pa and MPa, life in seconds and hours. This general guide covers alternatives; the report at the end of the interactive page contains only the selected current variant.

1. Task and inputs

First define the working-machine load and input method. Notation and initial relations follow below. Construction sizing runs in a background worker; parameter changes cancel stale requests. If a restored variant needs larger shafts and bearings, compatible dimensions are coordinated automatically. The initial variant becomes the comparison baseline if none exists; load, material and allowable stresses remain unchanged. Reload restores the saved variant. “Open verified example” first downloads a backup JSON. Save the task before changing construction.

Required torque T_3' and speed n_3' of the working shaft define required power:

$$P_3' = 2\pi T_3' n_3' \tag{1.1}$$

For tangential-force input F and linear speed v use the working diameter D of the driven element:

$$P_3' = F v \tag{1.2}$$

$$n_3' = \frac{v}{\pi D} \tag{1.3}$$

Equations use s^{-1}; fields use min^{-1}. Permissible relative speed deviation is denoted ε; actual deviation — δ_n. For a tolerance of 5%, the permitted range is −5% to +5%. Dimensions, efficiencies and safety factors are adopted for the task. Stock material and SKF bearings can be selected from the shared catalogues. Do not apply a safety factor again to an already allowable stress. Motor power is its mechanical shaft power. Displayed numbers use five significant figures; calculations and saved data retain full precision.

Shaft, gear and ratio notation

Shafts are numbered from the gearbox input. On the intermediate shaft, the second index distinguishes the first-stage wheel from the second-stage pinion. For planetary stages, s denotes the sun, p the planet and r the fixed ring; w denotes the number of worm starts. A prime denotes a desired value, for example u_1'; without a prime — the adopted value u_1. Iterations are numbered separately.

Table 1.1 — Shafts and quantity notation

Shaft	Gear elements	Power	Speed	Torque
1 · input	z_1	P_1	n_1	T_1
2 · intermediate	$z_{2\text{-}1}$; $z_{2\text{-}2}$	P_2	n_2	T_2
3 · output	z_3	P_3	n_3	T_3

u — gearbox reduction ratio, u_e — for the belt or chain, u_Σ — overall drive ratio; m denotes the motor:

$$u = u_1\, u_2 = \frac{n_1}{n_3} \tag{1.4}$$

$$u_\Sigma = u_\text{e} u = \frac{n_\text{m}}{n_3} \tag{1.5}$$

Table 1.2 — Parameters and adoption

Parameter	Teaching value / alternatives
Load input method q_{task}	Working-shaft torque and speed; Tangential force and linear speed
Required working-shaft torque T_3'	200 N·m Torque required by a mixer, machine tool, pump, conveyor or other working machine.
Required working-shaft speed n_3'	76,394 min⁻¹ Specified output shaft operating speed. Calculations use revolutions per second.
Working-machine tangential force F	2 000 N Tangential force at the driven element working circle; required torque T = FD/2.

Parameter	Teaching value / alternatives
Required linear speed v	0,8 m/s Linear speed at a wheel, pulley or drum working circle: v = πDn.
Driven element working diameter D	200 mm Working diameter at which tangential force F acts, such as a wheel, pulley or drum diameter; n = v/(πD), T = FD/2.
Gearbox type $q_{gearbox}$	Cylindrical; Combined; Bevel; Worm; Planetary The selection sets teaching defaults for the type. For a combined gearbox, select each stage separately. Type, geometry and parameters are saved in the project and determine the calculations and 3D model.
Stage 1: transmission type q_{stage1}	Cylindrical; Bevel; Worm; Planetary
Stage 1: tooth helix angle β β_1	0 ° β = 0° means spur teeth; β > 0° means helical teeth. External mating gears have opposite helix hands. Normal module m_n is set in section 4. Axial load increases with β; check bearings. This model range: 0…30°.
Stage 1: shaft angle Σ Σ_1	90 ° Straight bevel pair; pitch cones share a common apex. Usually Σ = 90°. Face width is measured along the cone generator.
Stage 1: number of worm starts $z_{w,1}$	2 Ratio u = z_2 / number of starts. Increasing starts changes lead angle and ratio.
Stage 1: worm diameter factor q q_{w1}	12 d_1 = q m_n; normal system compatible with BOSL2. Lead angle γ = arcsin(starts/q). Do not mix normal and axial modules.
Stage 1: planet tooth count $z_{p,1}$	27 Sun, planet and matching internal ring tooth counts are explained in section 2. The ring is fixed and the carrier is the output. Equal-spacing assembly and planet clearance are checked.
Stage 1: number of planets Np_1	3
Stage 2: transmission type q_{stage2}	Cylindrical; Bevel; Worm; Planetary

Parameter	Teaching value / alternatives
Stage 2: tooth helix angle β β_2	0 ° β = 0° means spur teeth; β > 0° means helical teeth. External mating gears have opposite helix hands. Normal module m_n is set in section 4. Axial load increases with β; check bearings. This model range: 0…30°.
Stage 2: shaft angle Σ Σ_2	90 ° Straight bevel pair; pitch cones share a common apex. Usually Σ = 90°. Face width is measured along the cone generator.
Stage 2: number of worm starts $z_{w,2}$	2 Ratio u = z_2 / number of starts. Increasing starts changes lead angle and ratio.
Stage 2: worm diameter factor q q_{w2}	12 $d_1 = q\ m_n$; normal system compatible with BOSL2. Lead angle γ = arcsin(starts/q). Do not mix normal and axial modules.
Stage 2: planet tooth count $z_{p,2}$	27 Sun, planet and matching internal ring tooth counts are explained in section 2. The ring is fixed and the carrier is the output. Equal-spacing assembly and planet clearance are checked.
Stage 2: number of planets Np_2	3
Gearbox stage count k	2
Input coupling efficiency η_c	0,99
Design load factor K	1,25
Permissible relative output speed deviation ε	5 % ε is the permissible relative speed deviation. The tolerance is symmetric ± the specified percentage. For example, 5 means −5% to +5%; shown in the checks table.
Belt type and standard section $q_{belt,profile}$	Custom profile / manual selection; Classical V-belt Z/10 · 10×6 mm; Classical V-belt A/13 · 13×8 mm; Classical V-belt B/17 · 17×11 mm; Classical V-belt C/22 · 22×14 mm; Classical V-belt D/32 · 32×20 mm; Classical V-belt E/40 · 40×25 mm; Narrow V-belt SPZ · 9,7×8 mm; Narrow V-belt SPA · 12,7×10 mm; Narrow V-belt SPB · 16,3×13 mm; Narrow V-belt SPC · 22×18 mm Classical Z/A/B/C/D/E or narrow SPZ/SPA/SPB/SPC V-belt. Nominal section and minimum pulley diameter are listed. Enter power rating per belt in section 3 for the specific product and speed.

2. Kinematics, power and torque

KNUBA guides: gearbox gear stage 3…6, chain 2…6, conventional belt up to 4, single-start worm 28…80, two-start worm 14…40 (p. 14, table 4). Extended teaching search ranges: cylindrical 2…8, bevel 2…5, planetary 3…8, four-start worm 8…20. Allocate more reduction to the worm stage. Automatic mode selects all stages; editing desired u or teeth enables manual mode. Recommendations and acceptance buttons are directly below the fields in section 2. Overall drive ratio is uΣ, gearbox ratio is u. With unequal modules, equal stage ratios do not mean equal diameters; equations (2.1)–(2.2) give the geometric guide. Interactive selection compares integer teeth and envelopes while preserving the task, motor, modules and widths. This is a teaching compactness criterion; strength and bearing checks must follow. Select one or two stages and the transmission before the gearbox. Adopt integer tooth counts, then check actual output speed within ±ε. If it fails, adjust teeth and repeat kinematic and load calculations before adopting dimensions. Resolve insufficient power or torque by selecting a motor for its duty.

Table 2.1 — Parameters and adoption

Parameter	**Teaching value / alternatives**
Motor mechanical power P_{m}	2,2 kW Mechanical shaft power. Select a rating from the ABB reference series or enter your own nameplate data.
Motor operating speed n_{m}	1 440 min^{-1} Nameplate speed under load. Select a variant for the adopted power or enter custom data.
Ratio allocation q_{split}	Automatic allocation by type and dimensions; Manual allocation; apply the proposal below Automatic: allocate all stages by dimensions and speed. Editing a desired ratio or tooth count enables manual mode.
Desired belt / chain ratio u'_{e}	2 Guide: conventional belt 2…4; chain 2…6. Timing belt search guide 2…6; check minimum teeth, wrap and product rating. These are not hard limits; editing enables manual mode.
Desired first-stage ratio u'_1	4 Planetary guide: 3…8, u = 1 + z_r/z_s with fixed ring and carrier output; tooth counts and planet count must permit assembly. Cylindrical stage: start with 3…6, extended search guide 2…8; bevel 2…5. Worm: single-start 28…80, two-start 14…40; four-start teaching guide 8…20. Overall drive ratio comes from motor and specified output speed and is allocated between the external transmission and gearbox. Similar ratios give similar diameters only with compatible modules, pinion tooth counts and helix angles; allocate more reduction to the worm stage. The recommendation below accounts for type and size; check modules, widths, strength and bearings after allocation.

For two cylindrical stages with similar modules, pinion tooth counts and helix angles, the initial allocation is:

$$u_1 \approx u_2 \approx \sqrt{u} \tag{2.1}$$

For different geometries, a guide to similar gear diameters is:

$$u_1 \approx \sqrt{u \frac{m_{n,2} z_{2\text{-}2} / \cos \beta_2}{m_{n,1} z_1 / \cos \beta_1}} \tag{2.2}$$

Required working-machine mechanical power uses the specified shaft torque and speed: P = Tω, where ω = 2πn. [6]

P_3' Working-machine power, kW.
T_3' Required working-shaft torque, N·m.
n_3' Required working-shaft speed, s^{-1}.
π number π, dimensionless.

$$P_3' = 2 \cdot \pi \cdot T_3' \cdot n_3' \tag{2.5}$$

$$P_3' = 2 \cdot 3{,}14159 \cdot 200 \cdot 1{,}27324 = 1600 \text{ W} = 1{,}6 \text{ kW} \tag{2.5a}$$

Take required working-shaft speed from the task. Multiply s^{-1} by 60 to obtain min^{-1}. [6]

n_3' Required working-shaft speed, min^{-1}.
n_3' Required working-shaft speed, s^{-1}.

$$n_3' = n_3' \tag{2.6}$$

$$n_3' = 1{,}2732 \text{ s}^{-1} = 76{,}394 \text{ min}^{-1} \tag{2.6a}$$

The required overall ratio is motor operating speed divided by required working-shaft speed. [6]

u_Σ' Required ratio, dimensionless.
n_m Motor operating speed, s^{-1}.
n_3' Required working-shaft speed, s^{-1}.

$$u'_{\Sigma} = \frac{n_{\mathrm{m}}}{n'_3} \tag{2.7}$$

$$u'_{\Sigma} = \frac{24}{1{,}27324} = 18{,}85 \tag{2.7a}$$

Adjust the final gear tooth count to match output speed. After rounding to an integer, repeat the kinematic calculation and specified tolerance check. [6]

z'_3 Refined final gear tooth count, dimensionless.

u_1 Stage 1: reduction ratio, dimensionless.

u_{e} Flexible transmission ratio, dimensionless.

u'_{Σ} Required ratio, dimensionless.

$z_{2\text{-}2}$ Stage 2: pinion tooth count, dimensionless.

$$z'_3 = \frac{u'_{\Sigma}}{(u_{\mathrm{e}} \cdot u_1)} \cdot z_{2\text{-}2} \tag{2.8}$$

$$z'_3 = \frac{18{,}8496}{(1 \cdot 5{,}5)} \cdot 20 = 68{,}544 \tag{2.8a}$$

Use the calculated ratio of the selected flexible transmission. Without a flexible transmission, use unity. [2]

u_{e} Flexible transmission ratio, dimensionless.

$$u_{\mathrm{e}} = 1 \tag{2.9}$$

$$u_{\mathrm{e}} = 1 \tag{2.9a}$$

Use the entered efficiency for the selected flexible transmission. Without one, omit this loss and use unity. [2]

η_{ext} Flexible transmission efficiency, dimensionless.

$$\eta_{ext} = 1 \tag{2.10}$$

$$\eta_{ext} = 1 \tag{2.10a}$$

Gearbox input shaft speed accounts for the flexible transmission ratio. [6]

n_1 Gearbox input speed, min^{-1}.
n_m Motor operating speed, s^{-1}.
u_e Flexible transmission ratio, dimensionless.

$$n_1 = \frac{n_\text{m}}{u_\text{e}} \tag{2.11}$$

$$n_1 = \frac{24}{1} = 24\ \text{s}^{-1} = 1440\ \text{min}^{-1} \tag{2.11a}$$

Gearbox input power decreases through input coupling and flexible transmission losses. [6]

P_1 Gearbox input power, kW.
P_m Motor mechanical power, W.
η_c Input coupling efficiency, dimensionless.
η_{ext} Flexible transmission efficiency, dimensionless.

$$P_1 = P_\text{m} \cdot \eta_c \cdot \eta_{ext} \tag{2.12}$$

$$P_1 = 2200 \cdot 0{,}99 \cdot 1 = 2178\ \text{W} = 2{,}178\ \text{kW} \tag{2.12a}$$

Torque follows from mechanical power and angular speed ω = 2πn. [6]

T_1 Gearbox input torque, N·m.
P_1 Gearbox input power, W.
n_1 Gearbox input speed, s^{-1}.
π number π, dimensionless.

$$T_1 = \frac{P_1}{(2 \cdot \pi \cdot n_1)} \tag{2.13}$$

$$T_1 = \frac{2178}{(2 \cdot 3{,}14159 \cdot 24)} = 14{,}443\ \text{N·m} \tag{2.13a}$$

For stage 1. Ratio follows from tooth counts for the adopted type: gear/pinion, wheel/worm starts, or 1 + ring/sun for a fixed ring. [2]

u_1 Stage 1: reduction ratio, dimensionless.
z_1 Stage 1: pinion tooth count, dimensionless.
$z_{2\text{-}1}$ Stage 1: wheel tooth count, dimensionless.

$$u_1 = \frac{z_{2\text{-}1}}{z_1} \tag{2.14}$$

$$u_1 = \frac{110}{20} = 5{,}5 \tag{2.14a}$$

For stage 1. Stage output speed equals input speed divided by ratio. [2]

n_2 Stage 1: output speed, min^{-1}.
n_1 Gearbox input speed, s^{-1}.
u_1 Stage 1: reduction ratio, dimensionless.

$$n_2 = \frac{n_1}{u_1} \tag{2.15}$$

$$n_2 = \frac{24}{5{,}5} = 4{,}3636\ \text{s}^{-1} = 261{,}82\ \text{min}^{-1} \tag{2.15a}$$

For stage 1. Stage output power includes its adopted efficiency. [2]

P_2 Stage 1: output power, kW.
P_1 Gearbox input power, W.
η_1 Stage 1: efficiency including losses, dimensionless.

$$P_2 = P_1 \cdot \eta_1 \tag{2.16}$$

$$P_2 = 2178 \cdot 0{,}98 = 2134{,}4\ \text{W} = 2{,}1344\ \text{kW} \tag{2.16a}$$

For stage 1. Stage output torque is determined from power and angular speed. [2]

T_2 Stage 1: output torque, N·m.
P_2 Stage 1: output power, W.
n_2 Stage 1: output speed, s^{-1}.
π number π, dimensionless.

$$T_2 = \frac{P_2}{(2 \cdot \pi \cdot n_2)} \tag{2.17}$$

$$T_2 = \frac{2134{,}44}{(2 \cdot 3{,}14159 \cdot 4{,}36364)} = 77{,}849 \text{ N·m} \tag{2.17a}$$

For stage 2. Ratio follows from tooth counts for the adopted type: gear/pinion, wheel/worm starts, or 1 + ring/sun for a fixed ring. [2]

u_2 Stage 2: reduction ratio, dimensionless.
$z_{2\text{-}2}$ Stage 2: pinion tooth count, dimensionless.
z_3 Stage 2: wheel tooth count, dimensionless.

$$u_2 = \frac{z_3}{z_{2\text{-}2}} \tag{2.18}$$

$$u_2 = \frac{69}{20} = 3{,}45 \tag{2.18a}$$

For stage 2. Stage output speed equals input speed divided by ratio. [2]

n_3 Stage 2: output speed, min^{-1}.
n_2 Stage 1: output speed, s^{-1}.
u_2 Stage 2: reduction ratio, dimensionless.

$$n_3 = \frac{n_2}{u_2} \tag{2.19}$$

$$n_3 = \frac{4{,}36364}{3{,}45} = 1{,}2648 \text{ s}^{-1} = 75{,}889 \text{ min}^{-1} \tag{2.19a}$$

For stage 2. Stage output power includes its adopted efficiency. [2]

P_3 Stage 2: output power, kW.
P_2 Stage 1: output power, W.
η_2 Stage 2: efficiency including losses, dimensionless.

$$P_3 = P_2 \cdot \eta_2 \tag{2.20}$$

$$P_3 = 2134{,}44 \cdot 0{,}98 = 2091{,}8\ \text{W} = 2{,}0918\ \text{kW} \tag{2.20a}$$

For stage 2. Stage output torque is determined from power and angular speed. [2]

T_3 Stage 2: output torque, N·m.
P_3 Stage 2: output power, W.
n_3 Stage 2: output speed, s^{-1}.
π number π, dimensionless.

$$T_3 = \frac{P_3}{(2 \cdot \pi \cdot n_3)} \tag{2.21}$$

$$T_3 = \frac{2091{,}75}{(2 \cdot 3{,}14159 \cdot 1{,}26482)} = 263{,}21\ \text{N·m} \tag{2.21a}$$

Output speed is taken from the last active gearbox stage; the stage count k is 1 or 2. [6]

n_3 Actual output speed, min^{-1}.
k Gearbox stage count, dimensionless.
n_2 Stage 1: output speed, s^{-1}.
n_3 Stage 2: output speed, s^{-1}.

$$n_3 = (2 - k) \cdot n_2 + (k - 1) \cdot n_3 \tag{2.22}$$

$$n_3 = (2 - 2) \cdot 4{,}36364 + (2 - 1) \cdot 1{,}26482 = 1{,}2648\ \text{s}^{-1} = 75{,}889\ \text{min}^{-1} \tag{2.22a}$$

Output power is taken from the last active gearbox stage after its losses. [6]

P_3 Actual output power, kW.
P_2 Stage 1: output power, W.
P_3 Stage 2: output power, W.

k Gearbox stage count, dimensionless.

$$P_3 = (2 - k) \cdot P_2 + (k - 1) \cdot P_3 \tag{2.23}$$

$$P_3 = (2 - 2) \cdot 2134{,}44 + (2 - 1) \cdot 2091{,}75 = 2091{,}8\ \text{W} = 2{,}0918\ \text{kW} \tag{2.23a}$$

Actual output torque follows from working-shaft power and speed. [6]

T_3 Actual output torque, N·m.
P_3 Actual output power, W.
n_3 Actual output speed, s^{-1}.
π number π, dimensionless.

$$T_3 = \frac{P_3}{(2 \cdot \pi \cdot n_3)} \tag{2.24}$$

$$T_3 = \frac{2091{,}75}{(2 \cdot 3{,}14159 \cdot 1{,}26482)} = 263{,}21\ \text{N·m} \tag{2.24a}$$

Calculate actual driven-element working-circle speed after adopting integer tooth counts. [6]

v_{out} Actual working-machine linear speed, m/s.
D Driven element working diameter, m.
n_3 Actual output speed, s^{-1}.
π number π, dimensionless.

$$v_{out} = \pi \cdot D \cdot n_3 \tag{2.25}$$

$$v_{out} = 3{,}14159 \cdot 0{,}2 \cdot 1{,}26482 = 0{,}79471\ \text{m/s} \tag{2.25a}$$

Signed output-speed deviation is determined after adopting integer tooth counts. Its absolute value is compared with the specified tolerance ±ε. [6]

δ_n Output-speed deviation, %.
n_3 Actual output speed, s^{-1}.
n_3' Required working-shaft speed, s^{-1}.

$$\delta_n = \frac{(n_3 - n_3')}{n_3'} \tag{2.26}$$

$$\delta_n = \frac{(1{,}26482 - 1{,}27324)}{1{,}27324} = -0{,}006611 = -0{,}6611 \tag{2.26a}$$

Overall efficiency is output mechanical power divided by motor power. [6]

η Overall drive efficiency, %.
P_{m} Motor mechanical power, W.
P_3 Actual output power, W.

$$\eta = \frac{P_3}{P_{\text{m}}} \tag{2.27}$$

$$\eta = \frac{2091{,}75}{2200} = 0{,}9508 = 95{,}08 \tag{2.27a}$$

Required motor mechanical power includes losses in all active drive components. [6]

P_{m}' Required motor power, kW.
P_3' Working-machine power, W.
η Overall drive efficiency, dimensionless.

$$P_{\text{m}}' = \frac{P_3'}{\eta} \tag{2.28}$$

$$P_{\text{m}}' = \frac{1600}{0{,}950796} = 1682{,}8\ \text{W} = 1{,}6828\ \text{kW} \tag{2.28a}$$

Check the power margin: adopted motor power divided by required power must be at least unity. [6]

r_P Available-to-required power ratio, dimensionless.
P_{m} Motor mechanical power, W.
P_{m}' Required motor power, W.

$$r_P = \frac{P_{\text{m}}}{P'_{\text{m}}} \tag{2.29}$$

$$r_P = \frac{2200}{1682{,}8} = 1{,}3073 \tag{2.29a}$$

Mechanical power loss is motor power minus output power. [6]

P_{loss} Power loss, W.
P_{m} Motor mechanical power, W.
P_3 Actual output power, W.

$$P_{loss} = P_{\text{m}} - P_3 \tag{2.30}$$

$$P_{loss} = 2200 - 2091{,}75 = 108{,}25\ \text{W} \tag{2.30a}$$

Recommended allocation

Table 2.2 — Recommended allocation: stage geometry and outputs

Stage / teeth	u_j	m_j, mm	b_j, mm	Envelope circle, mm	Centre / cone distance, mm	P_{j+1}, kW	n_{j+1}, min⁻¹	T_{j+1}, N·m
1 · Cylindrical $z_1 = 20$; $z_{2\text{-}1} = 110$	5,5	2,5	30	275	162,5	2,1344	261,82	77,84
2 · Cylindrical $z_{2\text{-}2} = 20$; $z_3 = 69$	3,45	4	40	276	178	2,0918	75,889	263,2

Table 2.3 — Recommended allocation: speed, power, torque and checks

Indicator	Symbol	Calculated	Specified / limit	Conclusion
External transmission ratio	u_{e}	1	—	—

Indicator	Symbol	Calculated	Specified / limit	Conclusion
Gearbox ratio	u	18,975	—	—
Overall drive ratio	u_{Σ}	18,975	$u'_{\Sigma} = 18,85$	—
Output speed, min⁻¹	n_3	75,889	$n'_3 = 76,394$	Satisfied
Relative speed deviation, %	δ_n	-0,6611	$\pm\varepsilon = \pm 5\%$	Satisfied
Required motor power, kW	P'_{m}	1,6828	$P_{\text{m}} = 2,2$	Satisfied
Output torque, N·m	T_3	263,21	$T'_3 = 200$	Satisfied
Summary of the listed checks	—	All listed conditions satisfied	See also Section 5	Satisfied

Manual allocation

Table 2.4 — Manual allocation: stage geometry and outputs

Stage / teeth	u_j	m_j, mm	b_j, mm	Envelope circle, mm	Centre / cone distance, mm	P_{j+1}, kW	n_{j+1}, min⁻¹	T_{j+1}, N·m
1 · Cylindrical $z_1 = 20$; $z_{2\text{-}1} = 80$	4	2,5	30	200	125	2,1344	360	56,618
2 · Cylindrical $z_{2\text{-}2} = 20$; $z_3 = 94$	4,7	4	40	376	228	2,0918	76,596	260,78

Table 2.5 — Manual allocation: speed, power, torque and checks

Indicator	Symbol	Calculated	Specified / limit	Conclusion
External transmission ratio	u_{e}	1	—	—
Gearbox ratio	u	18,8	—	—
Overall drive ratio	u_{Σ}	18,8	$u'_{\Sigma} = 18,85$	—

Indicator	Symbol	Calculated	Specified / limit	Conclusion
Output speed, min⁻¹	n_3	76,596	$n_3' = 76,394$	Satisfied
Relative speed deviation, %	δ_n	0,2636	$\pm\varepsilon = \pm 5\%$	Satisfied
Required motor power, kW	P'_{m}	1,6828	$P_{\text{m}} = 2,2$	Satisfied
Output torque, N·m	T_3	260,78	$T_3' = 200$	Satisfied
Summary of the listed checks	—	All listed conditions satisfied	See also Section 5	Satisfied

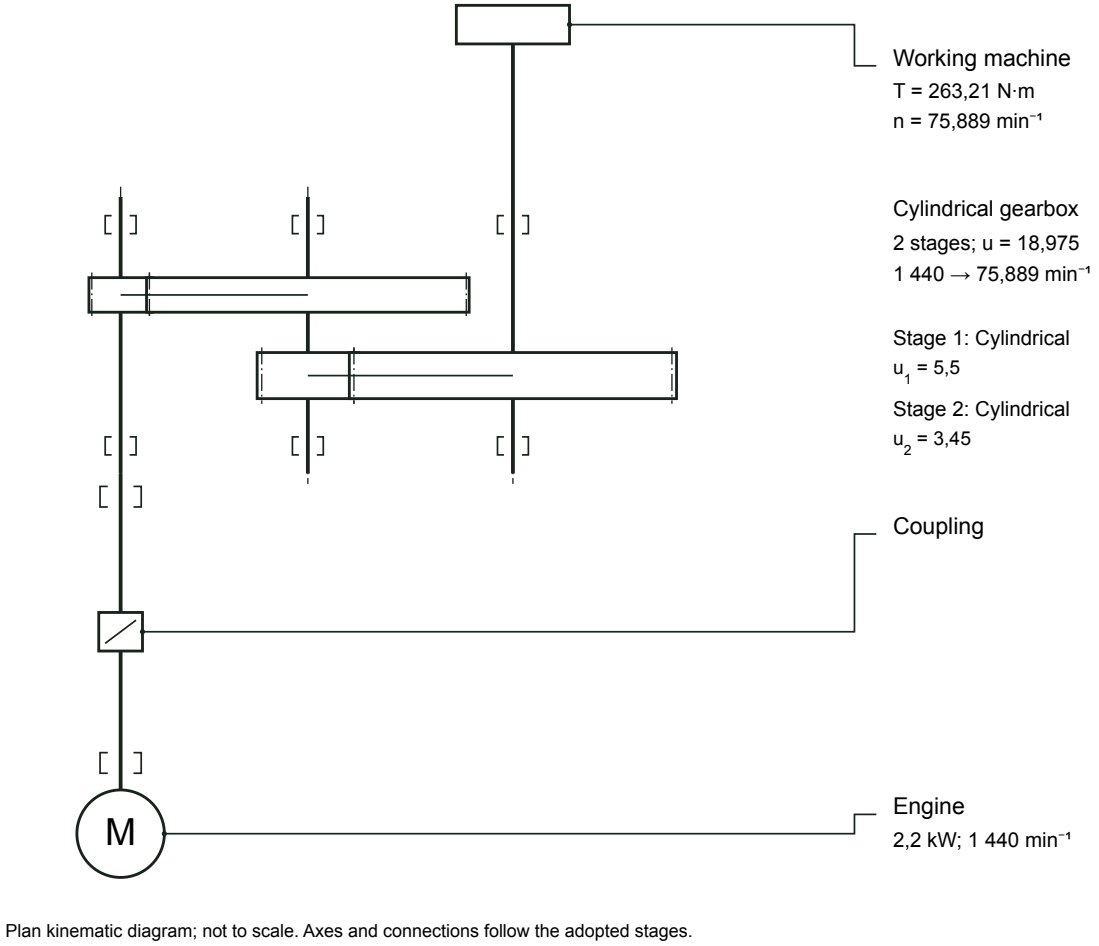


Figure 2.3 — Complete adopted drive diagram.

3. Belts, timing belts and chains

For a V-belt drive, select a classical Z/A/B/C/D/E or narrow SPZ/SPA/SPB/SPC section from the shared Optibelt CSV. Check minimum pulley datum diameters; do not mix Li and Ld designations. Use the compactness recommendation in section 2 for ratio

allocation, then check loads and bearings. The “No flexible transmission” option omits this section from the interactive calculation and report: $u_e = \eta_e = 1$. A conventional belt includes slip and span tensions; a timing belt uses a compatible profile, pitch, width and tooth count; a chain uses a compatible pitch, integer sprocket tooth counts and an even number of links. All three alternatives appear below. Enter belt or chain working ratings for the specific product; breaking force is not a permissible working load.

Table 3.1 — Parameters and adoption

Parameter	**Teaching value / alternatives**
Driving pulley datum diameter d_{b1}	100 mm Diameter d_d at the datum line, not the outside diameter. The minimum depends on the selected belt section.
Driven pulley datum diameter d_{b2}	200 mm Datum diameter d_d; must match the groove profile. Ratio includes the adopted slip.
Conventional belt slip s	2 %
Belt drive centre distance a_b	400 mm
Permissible power per belt P_b	1 kW For the selected profile, diameter and speed; the initial value is illustrative.
Wrap correction c_a	0,9
Length correction c_l	1
Preload per belt F_{0b}	200 N
Belt drive efficiency η_b	0,95

Parameter	Teaching value / alternatives
Standard timing belt $q_{timing,catalogue}$	Custom belt / manual data; Gates PowerGrip GT4 1120-8MGT-30; 140 teeth; Gates PowerGrip GT3 400-5MGT-15; 80 teeth; Gates PowerGrip GT3 500-5MGT-15; 100 teeth; Gates PowerGrip GT3 600-5MGT-15; 120 teeth; Gates PowerGrip GT3 700-5MGT-15; 140 teeth; Gates PowerGrip GT3 800-5MGT-15; 160 teeth; Gates PowerGrip GT3 1000-5MGT-15; 200 teeth; Gates PowerGrip GT3 1600-5MGT-15; 320 teeth; Gates PowerGrip GT4 600-8MGT-30; 75 teeth; Gates PowerGrip GT4 800-8MGT-30; 100 teeth; Gates PowerGrip GT4 960-8MGT-30; 120 teeth; Gates PowerGrip GT4 1280-8MGT-30; 160 teeth; Gates PowerGrip GT4 1440-8MGT-30; 180 teeth; Gates PowerGrip GT4 1600-8MGT-30; 200 teeth; Gates PowerGrip GT4 1400-14MGT-40; 100 teeth; Gates PowerGrip GT4 1400-14MGT-55; 100 teeth; Gates PowerGrip HTD 400-5M-15; 80 teeth; Gates PowerGrip HTD 500-5M-15; 100 teeth; Gates PowerGrip HTD 600-5M-15; 120 teeth; Gates PowerGrip HTD 800-5M-15; 160 teeth; Gates PowerGrip HTD 1000-5M-15; 200 teeth; Gates PowerGrip HTD 1000-8M-30; 125 teeth
Timing belt width b_{timing}	30 mm Catalogue width for the selected product. Working capacity comes from the manufacturer's rating with corrections; width alone does not define permissible force.
Timing belt pitch p_t	8 mm
Driving pulley teeth z_{t1}	24
Driven pulley teeth z_{t2}	48
Adopted belt tooth count Z_b	140
Permissible effective belt force F_{tlim}	600 N For the specific profile, width, speed and number of teeth in mesh.
Timing belt efficiency η_t	0,97
Standard roller chain $q_{chain,catalogue}$	Custom chain / manual data; Renold 05B-1; p = 8 mm; Renold 06B-1; p = 9.525 mm; Renold 08B-1; p = 12.7 mm; Renold 10B-1; p = 15.875 mm; Renold 12B-1; p = 19.05 mm; Renold 16B-1; p = 25.4 mm; Renold 20B-1; p = 31.75 mm; Renold 24B-1; p = 38.1 mm
Chain pitch p_c	12,7 mm

Parameter	Teaching value / alternatives
Driving sprocket teeth z_{c1}	19
Driven sprocket teeth z_{c2}	38
Adopted chain link count X_c	94
Permissible chain working force F_{clim}	3 000 N Working capacity for operating conditions; not breaking force.
Chain drive efficiency η_{ch}	0,96

Belt drive

Conventional belt ratio uses pulley diameters and adopted slip. [2]

u_b Conventional belt ratio, dimensionless.
d_{b1} Driving pulley datum diameter, m.
d_{b2} Driven pulley datum diameter, m.
s Conventional belt slip, dimensionless.

$$u_b = \frac{d_{b2}}{(d_{b1} \cdot (1 - s))} \tag{3.1}$$

$$u_b = \frac{0{,}196}{(0{,}1 \cdot (1 - 0{,}02))} = 2 \tag{3.1a}$$

Belt speed uses the driving pulley rotating at motor speed. [2]

v_b Belt speed, m/s.
d_{b1} Driving pulley datum diameter, m.
n_{m} Motor operating speed, s^{-1}.
π number π, dimensionless.

$$v_b = \pi \cdot d_{b1} \cdot n_{\mathrm{m}} \tag{3.2}$$

$$v_b = 3{,}14159 \cdot 0{,}1 \cdot 24 = 7{,}5398\ \mathrm{m/s} \tag{3.2a}$$

Small pulley wrap angle uses an open-belt drive with parallel shafts. [2]

β Small pulley wrap angle, °.
a_b Belt drive centre distance, m.
d_{b1} Driving pulley datum diameter, m.
d_{b2} Driven pulley datum diameter, m.
π number π, dimensionless.

$$\beta = \pi - 2 \cdot \sin^{-1}\left(\frac{(d_{b2} - d_{b1})}{(2 \cdot a_b)}\right) \tag{3.3}$$

$$\beta = 3{,}14159 - 2 \cdot \sin^{-1}\left(\frac{(0{,}196 - 0{,}1)}{(2 \cdot 0{,}4)}\right) = 2{,}901 = 166{,}22\ ^\circ \tag{3.3a}$$

Estimate open-belt length using an approximate geometric relation; select the catalogue length separately. [2]

L_b Calculated belt length, mm.
a_b Belt drive centre distance, m.
d_{b1} Driving pulley datum diameter, m.
d_{b2} Driven pulley datum diameter, m.
π number π, dimensionless.

$$L_b = 2 \cdot a_b + \frac{\pi \cdot (d_{b1} + d_{b2})}{2} + \frac{(d_{b2} - d_{b1})^2}{(4 \cdot a_b)} \tag{3.4}$$

$$L_b = 2 \cdot 0{,}4 + \frac{3{,}14159 \cdot (0{,}1 + 0{,}196)}{2} + \frac{(0{,}196 - 0{,}1)^2}{(4 \cdot 0{,}4)} = 1{,}2707\ \mathrm{m} = 1270{,}7\ \mathrm{mm} \tag{3.4a}$$

Round the required belt count upward after load and catalogue power corrections per belt. [2]

N_b Required belt count, dimensionless.
K Design load factor, dimensionless.
P_b Permissible power per belt, W.
P_{m} Motor mechanical power, W.

c_a Wrap correction, dimensionless.
c_l Length correction, dimensionless.

$$N_b = \left\lceil \frac{P_{\mathrm{m}} \cdot K}{(P_b \cdot c_a \cdot c_l)} \right\rceil \tag{3.5}$$

$$N_b = \left\lceil \frac{2200 \cdot 1{,}25}{(1000 \cdot 0{,}9 \cdot 1)} \right\rceil = 4 \tag{3.5a}$$

Effective belt force follows from transmitted power and belt speed. [2]

F_b Effective tangential force, N.
P_{m} Motor mechanical power, W.
v_b Belt speed, m/s.

$$F_b = \frac{P_{\mathrm{m}}}{v_b} \tag{3.6}$$

$$F_b = \frac{2200}{7{,}53982} = 291{,}78\ \mathrm{N} \tag{3.6a}$$

Tight-side tension per belt follows from preload and its share of design tangential force. [2]

F_1 Tight-side tension per belt, N.
F_{0b} Preload per belt, N.
F_b Effective tangential force, N.
K Design load factor, dimensionless.
N_b Required belt count, dimensionless.

$$F_1 = F_{0b} + \frac{K \cdot F_b}{(2 \cdot N_b)} \tag{3.7}$$

$$F_1 = 200 + \frac{1{,}25 \cdot 291{,}784}{(2 \cdot 4)} = 245{,}59\ \mathrm{N} \tag{3.7a}$$

Slack-side tension must remain positive; centrifugal force is excluded from this teaching model. [2]

F_2 Slack-side tension per belt, N.
F_{0b} Preload per belt, N.
F_b Effective tangential force, N.
K Design load factor, dimensionless.
N_b Required belt count, dimensionless.

$$F_2 = F_{0b} - \frac{K \cdot F_b}{(2 \cdot N_b)} \tag{3.8}$$

$$F_2 = 200 - \frac{1{,}25 \cdot 291{,}784}{(2 \cdot 4)} = 154{,}41 \text{ N} \tag{3.8a}$$

Estimate pulley shaft load from span tensions and wrap angle. Check the input shaft separately. [2]

F_{shaft} Approximate pulley shaft load, N.
F_{0b} Preload per belt, N.
N_b Required belt count, dimensionless.
β Small pulley wrap angle, dimensionless.

$$F_{shaft} = 2 \cdot F_{0b} \cdot N_b \cdot \sin\left(\frac{\beta}{2}\right) \tag{3.9}$$

$$F_{shaft} = 2 \cdot 200 \cdot 4 \cdot \sin\left(\frac{2{,}90101}{2}\right) = 1588{,}4 \text{ N} \tag{3.9a}$$

Timing belt

Timing belt ratio uses integer pulley tooth counts without slip. [31]

u_t Timing belt ratio, dimensionless.
z_{t1} Driving pulley teeth, dimensionless.
z_{t2} Driven pulley teeth, dimensionless.

$$u_t = \frac{z_{t2}}{z_{t1}} \tag{3.10}$$

$$u_t = \frac{64}{24} = 2{,}6667 \tag{3.10a}$$

The driving timing pulley pitch diameter follows from its pitch and tooth count. [31]

d_{t1} Small pulley pitch diameter, mm.
p_t Timing belt pitch, m.
π number π, dimensionless.
z_{t1} Driving pulley teeth, dimensionless.

$$d_{t1} = \frac{p_t \cdot z_{t1}}{\pi} \tag{3.11}$$

$$d_{t1} = \frac{0{,}008 \cdot 24}{3{,}14159} = 0{,}061115\ \text{m} = 61{,}115\ \text{mm} \tag{3.11a}$$

The driven timing pulley pitch diameter uses the same pitch and its tooth count. [31]

d_{t2} Large pulley pitch diameter, mm.
p_t Timing belt pitch, m.
π number π, dimensionless.
z_{t2} Driven pulley teeth, dimensionless.

$$d_{t2} = \frac{p_t \cdot z_{t2}}{\pi} \tag{3.12}$$

$$d_{t2} = \frac{0{,}008 \cdot 64}{3{,}14159} = 0{,}16297\ \text{m} = 162{,}97\ \text{mm} \tag{3.12a}$$

Timing belt length uses an integer tooth count and the pitch of a compatible profile. [31]

L_t Adopted timing belt length, mm.
Z_b Adopted belt tooth count, dimensionless.
p_t Timing belt pitch, m.

$$L_t = p_t \cdot Z_b \tag{3.13}$$

$$L_t = 0{,}008 \cdot 140 = 1{,}12\ \text{m} = 1120\ \text{mm} \tag{3.13a}$$

Refine centre distance using the adopted belt length; the radicand must be positive. [31]

a_t Adopted belt centre distance, mm.
A_t Auxiliary length quantity, m.
d_{t1} Small pulley pitch diameter, m.
d_{t2} Large pulley pitch diameter, m.

$$a_t = \frac{\left(A_t + \sqrt{{A_t}^2 - 2 \cdot (d_{t2} - d_{t1})^2}\right)}{4} \tag{3.14}$$

$$a_t = \frac{\left(0{,}768 + \sqrt{0{,}768^2 - 2 \cdot (0{,}162975 - 0{,}0611155)^2}\right)}{4} = 0{,}38059\ \text{m} = 380{,}59\ \text{mm} \tag{3.14a}$$

Small timing pulley wrap angle uses the refined centre distance. [31]

β_t Small timing pulley wrap angle, °.
a_t Adopted belt centre distance, m.
d_{t1} Small pulley pitch diameter, m.
d_{t2} Large pulley pitch diameter, m.
π number π, dimensionless.

$$\beta_t = \pi - 2 \cdot \sin^{-1}\left(\frac{(d_{t2} - d_{t1})}{(2 \cdot a_t)}\right) \tag{3.15}$$

$$\beta_t = 3{,}14159 - 2 \cdot \sin^{-1}\left(\frac{(0{,}162975 - 0{,}0611155)}{(2 \cdot 0{,}380592)}\right) = 2{,}8732 = 164{,}62\ ^\circ \tag{3.15a}$$

Teeth in mesh follow from the wrap angle. Check the permitted number against manufacturer requirements. [31]

Z_{mesh} Teeth in mesh, dimensionless.
β_t Small timing pulley wrap angle, dimensionless.
π number π, dimensionless.
z_{t1} Driving pulley teeth, dimensionless.

$$Z_{mesh} = \frac{z_{t1} \cdot \beta_t}{(2 \cdot \pi)} \tag{3.16}$$

$$Z_{mesh} = \frac{24 \cdot 2{,}87315}{(2 \cdot 3{,}14159)} = 10{,}975 \tag{3.16a}$$

Mean timing belt speed follows from pitch, driving pulley tooth count and speed. [31]

v_t Timing belt speed, m/s.
n_{m} Motor operating speed, s^{-1}.
p_t Timing belt pitch, m.
z_{t1} Driving pulley teeth, dimensionless.

$$v_t = p_t \cdot z_{t1} \cdot n_{\mathrm{m}} \tag{3.17}$$

$$v_t = 0{,}008 \cdot 24 \cdot 24 = 4{,}608 \ \mathrm{m/s} \tag{3.17a}$$

Timing belt design force includes the load factor. [31]

F_t Timing belt design force, N.
K Design load factor, dimensionless.
P_{m} Motor mechanical power, W.
v_t Timing belt speed, m/s.

$$F_t = \frac{K \cdot P_{\mathrm{m}}}{v_t} \tag{3.18}$$

$$F_t = \frac{1{,}25 \cdot 2200}{4{,}608} = 596{,}79 \ \mathrm{N} \tag{3.18a}$$

Compare force with the permissible value for the specific belt profile and width; the ratio must not exceed unity. [31]

q_t Belt allowable-force utilization, dimensionless.
F_t Timing belt design force, N.
F_{tlim} Permissible effective belt force, N.

$$q_t = \frac{F_t}{F_{tlim}} \tag{3.19}$$

$$q_t = \frac{596{,}788}{600} = 0{,}99465 \tag{3.19a}$$

Chain drive

Chain drive ratio follows from sprocket tooth counts. [32]

u_c Chain ratio, dimensionless.
z_{c1} Driving sprocket teeth, dimensionless.
z_{c2} Driven sprocket teeth, dimensionless.

$$u_c = \frac{z_{c2}}{z_{c1}} \tag{3.20}$$

$$u_c = \frac{44}{19} = 2{,}3158 \tag{3.20a}$$

The driving sprocket pitch diameter follows from the chain pitch and tooth count. [32]

d_{c1} Small sprocket pitch diameter, mm.
p_c Chain pitch, m.
π number π, dimensionless.
z_{c1} Driving sprocket teeth, dimensionless.

$$d_{c1} = \frac{p_c}{\sin\left(\frac{\pi}{z_{c1}}\right)} \tag{3.21}$$

$$d_{c1} = \frac{0{,}0127}{\sin\left(\frac{3{,}14159}{19}\right)} = 0{,}077159\ \text{m} = 77{,}159\ \text{mm} \tag{3.21a}$$

The driven sprocket pitch diameter uses the same pitch and its tooth count. [32]

d_{c2} Large sprocket pitch diameter, mm.

p_c Chain pitch, m.
π number π, dimensionless.
z_{c2} Driven sprocket teeth, dimensionless.

$$d_{c2} = \frac{p_c}{\sin\left(\frac{\pi}{z_{c2}}\right)} \tag{3.22}$$

$$d_{c2} = \frac{0{,}0127}{\sin\left(\frac{3{,}14159}{44}\right)} = 0{,}17802\ \text{m} = 178{,}02\ \text{mm} \tag{3.22a}$$

Refine centre distance using the adopted even link count; the geometric relation is approximate. [32]

a_c Chain centre distance, mm.
A_c Auxiliary chain quantity, dimensionless.
p_c Chain pitch, m.
π number π, dimensionless.
z_{c1} Driving sprocket teeth, dimensionless.
z_{c2} Driven sprocket teeth, dimensionless.

$$a_c = \frac{p_c \cdot \left(A_c + \sqrt{{A_c}^2 - 2 \cdot \left(\frac{(z_{c2} - z_{c1})}{\pi}\right)^2}\right)}{4} \tag{3.23}$$

$$a_c = \frac{0{,}0127 \cdot \left(62{,}5 + \sqrt{62{,}5^2 - 2 \cdot \left(\frac{(44 - 19)}{3{,}14159}\right)^2}\right)}{4} = 0{,}39363\ \text{m} = 393{,}63\ \text{mm} \tag{3.23a}$$

Adopted chain length follows from link count and pitch. [32]

L_c Adopted chain length, mm.
X_c Adopted chain link count, dimensionless.
p_c Chain pitch, m.

$$L_c = p_c \cdot X_c \tag{3.24}$$

$$L_c = 0{,}0127 \cdot 94 = 1{,}1938\ \text{m} = 1193{,}8\ \text{mm} \tag{3.24a}$$

Mean chain speed follows from pitch, driving sprocket tooth count and speed. [32]

v_c Mean chain speed, m/s.
n_m Motor operating speed, s^{-1}.
p_c Chain pitch, m.
z_{c1} Driving sprocket teeth, dimensionless.

$$v_c = p_c \cdot z_{c1} \cdot n_\text{m} \tag{3.25}$$

$$v_c = 0{,}0127 \cdot 19 \cdot 24 = 5{,}7912\ \text{m/s} \tag{3.25a}$$

Chain working force includes the load factor. [32]

F_c Chain design working force, N.
K Design load factor, dimensionless.
P_m Motor mechanical power, W.
v_c Mean chain speed, m/s.

$$F_c = \frac{K \cdot P_\text{m}}{v_c} \tag{3.26}$$

$$F_c = \frac{1{,}25 \cdot 2200}{5{,}7912} = 474{,}86\ \text{N} \tag{3.26a}$$

Compare working force with the permissible operating value. Breaking force is not a permissible working load. [32]

q_c Chain allowable-force utilization, dimensionless.
F_c Chain design working force, N.
F_{clim} Permissible chain working force, N.

$$q_c = \frac{F_c}{F_{clim}} \tag{3.27}$$

$$q_c = \frac{474{,}858}{3000} = 0{,}15829 \tag{3.27a}$$

4. Gear drives

Select the gearbox type in section 1. A cylindrical stage has $\beta = 0°$ for spur teeth or $\beta > 0°$ for helical teeth; mating gears have opposite helix hands. Specify shaft angle for bevel gears, starts and q in the normal system for worm gears, and sun, planets and planet count for planetary gears. Combined permits different stage types. Selection determines kinematics and geometry; noncylindrical product strength is checked against its stated torque rating. Planetary stage: fixed ring, sun input, carrier output; $u_j = 1 + \frac{z_{r,j}}{z_{s,j}}$. Conditions: $z_{r,j} = z_{s,j} + 2z_{p,j}$; $\frac{z_{s,j}+z_{r,j}}{N_j}$ is an integer; $z_{p,j} + 2 < (z_{s,j} + z_{p,j})\sin(\pi/N_j)$. Size comparison uses the ring outer envelope including internal tooth roots and rim thickness. Select a module from the first reference series; second-series or custom values require tooling and availability justification. This model uses the conservative limit z ≥ 18. Index z_{1-1} denotes the first-stage pinion and z_{2-2} the second-stage gear. Involute flanks and face width form the 3D view geometry. The root transition fillet is approximate; the view is not a manufacturing drawing.

Select teaching assumptions for material, heat treatment, fits, accuracy grade, surface texture, geometric tolerances and hubs in section 4. Hub dimensions, induction hardening depth and tooth thickness reduction are in the collapsed fields. These requirements are saved in JSON and do not change strength calculation allowables. Gear datum A is the bore axis and B is the hub abutment face; shaft journal axes form common datum A–B. Ra, runout and perpendicularity are adopted manufacturing requirements. Cylindrical gears reference ISO 1328-1, bevel gears ISO 17485 and worm gears DIN 3974. Construction conflicts are shown before export; the nominal outline is not a toolpath.

Table 4.1 — Parameters and adoption

Parameter	Teaching value / alternatives
Stage 1: worm threaded length L_{w1}	80 mm
Stage 1: internal ring rim thickness t_{ring1}	6 mm Thickness from the root circle to the ring outside outline; at least one module. Check rim strength separately.
Stage 2: worm threaded length L_{w2}	80 mm
Stage 2: internal ring rim thickness t_{ring2}	6 mm Thickness from the root circle to the ring outside outline; at least one module. Check rim strength separately.

Parameter	Teaching value / alternatives
Stage 1: pinion tooth count z_1	20 Sun gear for a planetary stage. Integer z ≥ 18 for α = 20°, x = 0 is a conservative undercut limit. Fewer teeth require another profile model.
Stage 1: wheel tooth count $z_{2\text{-}1}$	80 Worm wheel for a worm stage. Integer z ≥ 18; check the ratio after rounding using the shaft notation in section 1.
Stage 1: module m_1	2,5 mm
Stage 1: face width b_1	30 mm
Stage 1: efficiency including losses η_1	0,98
Stage 2: pinion tooth count $z_{2\text{-}2}$	20 Sun gear for a planetary stage. Integer z ≥ 18 for α = 20°, x = 0 is a conservative undercut limit. Fewer teeth require another profile model.
Stage 2: wheel tooth count z_3	94 Worm wheel for a worm stage. Integer z ≥ 18; check the ratio after rounding using the shaft notation in section 1.
Stage 2: module m_2	4 mm
Stage 2: face width b_2	40 mm
Stage 2: efficiency including losses η_2	0,98
Gear-to-housing clearance c_{case}	12 mm Editable teaching housing dimension for A1; not a stiffness or thermal check.
Housing wall and cover thickness t_{case}	8 mm Teaching construction dimension. Housing walls are shown as hatched schematic sections.
Additional housing base thickness t_{base}	12 mm
Oil level above the internal floor h_{oil}	25 mm Adopted teaching oil level for the illustrated orientation. Refine level, oil grade and lubrication of all stages for the construction and duty.

Parameter	Teaching value / alternatives
Gear material and heat treatment mat_{doc}	C45 +N · 170–210 HBW; C45 · induction-hardened teeth 50–60 HRC; 42CrMo4 +QT · 270–320 HBW Teaching drawing requirements. The worm wheel uses CAC502 bronze without hardening. These requirements do not change section 5 allowable stresses.
Gear-to-shaft fit fit_{doc}	H7/k6 · transition, keyed; H7/h6 · clearance, keyed Teaching fit selection to ISO 286. The bore and matching shaft section receive compatible designations and numerical deviations.
Tooth accuracy grade Q_{doc}	8 ISO 1328-1 for cylindrical, ISO 17485 for bevel; DIN 3974 for worm gears. Grade is an adopted manufacturing requirement.
Surface finish Ra_{doc}	Ra: bore 1.6; teeth 3.2; others 6.3 μm; Ra: bore 0.8; teeth 1.6; others 3.2 μm
Geometrical tolerances gdt_{doc}	Runout 0.03; perpendicularity 0.02 mm; Runout 0.02; perpendicularity 0.01 mm Teaching requirements relative to bore datum A and abutment face B; shafts use common journal datum A–B. ISO 1101.
Gear hub configuration hub_{doc}	Disc without an extended hub; One-sided hub Construction dimensions for drawings. The worm is integral with its shaft; planets and internal ring have no projecting hub.
Hub diameter / bore diameter $Dhub_d$	1,5 Teaching proportion, not a hub strength calculation result.
Hub extension / face width $Lhub_b$	0,5
Effective induction hardening depth to 450 HV h_{eff}	1 mm Adopted manufacturing requirement for induction hardening; inspect zone and hardness after treatment.
Normal tooth thickness reduction Asn_{doc}	0,04 mm Adopted reduction on each gear. Deviation ±0.01 mm around this reduction; nominal 3D outlines are not toolpaths.

For stage 1. Transverse pressure angle follows from the normal pressure angle and helix angle β. [20]

α_{t1} Stage 1: transverse pressure angle, °.

α tooth pressure angle; adopted 20°, rad.

β_1 Stage 1: tooth helix angle β, dimensionless.

$$\alpha_{t1} = \tan^{-1}\left(\frac{\tan(\alpha)}{\cos(\beta_1)}\right) \tag{4.1}$$

$$\alpha_{t1} = \tan^{-1}\left(\frac{\tan(0{,}349066)}{\cos(0)}\right) = 0{,}34907 = 20\ ^\circ \tag{4.1a}$$

For stage 1. The worm pitch lead angle follows from its number of starts and diameter factor in the normal-module system. [20]

γ_1 Stage 1: worm lead angle, °.

q_{w1} Stage 1: worm diameter factor q, dimensionless.

$z_{\mathrm{w},1}$ Stage 1: number of worm starts, dimensionless.

$$\gamma_1 = \sin^{-1}\left(\frac{z_{\mathrm{w},1}}{q_{w1}}\right) \tag{4.2}$$

$$\gamma_1 = \sin^{-1}\left(\frac{2}{12}\right) = 0{,}16745 = 9{,}5941\ ^\circ \tag{4.2a}$$

For stage 1. For a simple planetary gear set, the ring tooth count equals the sun tooth count plus twice the planet tooth count. [20]

$z_{\mathrm{r},1}$ Stage 1: internal ring tooth count, dimensionless.

$z_{\mathrm{s},1}$ Stage 1: pinion tooth count, dimensionless.

$z_{\mathrm{p},1}$ Stage 1: planet tooth count, dimensionless.

$$z_{\mathrm{r},1} = z_{\mathrm{s},1} + 2 \cdot z_{\mathrm{p},1} \tag{4.3}$$

$$z_{\mathrm{r},1} = 24 + 2 \cdot 27 = 78 \tag{4.3a}$$

For stage 1. The pinion pitch cone angle follows from ratio and specified shaft angle for an external bevel pair. [20]

δ_1 Stage 1: pinion pitch cone angle, °.

Σ_1 Stage 1: shaft angle Σ, dimensionless.

z_1 Stage 1: pinion tooth count, dimensionless.

$z_{2\text{-}1}$ Stage 1: wheel tooth count, dimensionless.

$$\delta_1 = \cos^{-1}\left(\frac{\left(\frac{z_{2\text{-}1}}{z_1} + \cos\left(\Sigma_1\right)\right)}{\sqrt{\left(\frac{z_{2\text{-}1}}{z_1}\right)^2 + \frac{2 \cdot z_{2\text{-}1}}{z_1} \cdot \cos\left(\Sigma_1\right) + 1}}\right) \tag{4.4}$$

$$\delta_1 = \cos^{-1}\left(\frac{\left(\frac{110}{20} + \cos\left(1{,}5708\right)\right)}{\sqrt{\left(\frac{110}{20}\right)^2 + \frac{2 \cdot 110}{20} \cdot \cos\left(1{,}5708\right) + 1}}\right) = 0{,}17985 = 10{,}305\ ^\circ \tag{4.4a}$$

For stage 1. The gear pitch cone angle equals the specified shaft angle minus the pinion cone angle. [20]

$\delta_{2\text{-}1}$ Stage 1: wheel pitch cone angle, °.
Σ_1 Stage 1: shaft angle Σ, dimensionless.
δ_1 Stage 1: pinion pitch cone angle, dimensionless.

$$\delta_{2\text{-}1} = \Sigma_1 - \delta_1 \tag{4.5}$$

$$\delta_{2\text{-}1} = 1{,}5708 - 0{,}179853 = 1{,}3909 = 79{,}695\ ^\circ \tag{4.5a}$$

For stage 1. Outer cone distance follows from the pinion pitch diameter and cone angle. Face width must be smaller than this distance. [20]

R_1 Stage 1: outer cone distance, mm.
δ_1 Stage 1: pinion pitch cone angle, dimensionless.
m_1 Stage 1: module, m.
z_1 Stage 1: pinion tooth count, dimensionless.

$$R_1 = \frac{m_1 \cdot z_1}{\left(2 \cdot \sin\left(\delta_1\right)\right)} \tag{4.6}$$

$$R_1 = \frac{0{,}0025 \cdot 20}{\left(2 \cdot \sin\left(0{,}179853\right)\right)} = 0{,}13975\ \text{m} = 139{,}75\ \text{mm} \tag{4.6a}$$

For stage 1. The planet pitch diameter follows from the module and integer tooth count. [20]

d_{p1} Stage 1: planet pitch diameter, mm.

m_1 Stage 1: module, m.

$z_{p,1}$ Stage 1: planet tooth count, dimensionless.

$$d_{p1} = m_1 \cdot z_{p,1} \tag{4.7}$$

$$d_{p1} = 0{,}0025 \cdot 27 = 0{,}0675\ \text{m} = 67{,}5\ \text{mm} \tag{4.7a}$$

For stage 1. The pinion pitch diameter follows from the normal module, tooth count and helix angle β: the transverse module is $m_n/\cos\beta$. [20]

d_1 Stage 1: pinion diameter, mm.

β_1 Stage 1: tooth helix angle β, dimensionless.

m_1 Stage 1: module, m.

z_1 Stage 1: pinion tooth count, dimensionless.

$$d_1 = \frac{m_1 \cdot z_1}{\cos(\beta_1)} \tag{4.8}$$

$$d_1 = \frac{0{,}0025 \cdot 20}{\cos(0)} = 0{,}05\ \text{m} = 50\ \text{mm} \tag{4.8a}$$

For stage 1. The gear pitch diameter uses the same normal module and helix angle β. [20]

$d_{2\text{-}1}$ Stage 1: wheel diameter, mm.

β_1 Stage 1: tooth helix angle β, dimensionless.

m_1 Stage 1: module, m.

$z_{2\text{-}1}$ Stage 1: wheel tooth count, dimensionless.

$$d_{2\text{-}1} = \frac{m_1 \cdot z_{2\text{-}1}}{\cos(\beta_1)} \tag{4.9}$$

$$d_{2\text{-}1} = \frac{0{,}0025 \cdot 110}{\cos(0)} = 0{,}275\ \text{m} = 275\ \text{mm} \tag{4.9a}$$

For stage 1. Centre distance depends on the adopted mesh type. Bevel gear axes intersect, so this distance is zero; cone distance is calculated separately. [20]

a_{w1} Stage 1: centre distance, mm.
d_1 Stage 1: pinion diameter, m.
$d_{2\text{-}1}$ Stage 1: wheel diameter, m.

$$a_{w1} = \frac{(d_1 + d_{2\text{-}1})}{2} \tag{4.10}$$

$$a_{w1} = \frac{(0{,}05 + 0{,}275)}{2} = 0{,}1625\ \text{m} = 162{,}5\ \text{mm} \tag{4.10a}$$

For stage 1. Pinion tip diameter uses an addendum equal to the module. [20]

$d_{a,1}$ Stage 1: pinion tip diameter, mm.
d_1 Stage 1: pinion diameter, m.
m_1 Stage 1: module, m.

$$d_{a,1} = d_1 + 2 \cdot m_1 \tag{4.11}$$

$$d_{a,1} = 0{,}05 + 2 \cdot 0{,}0025 = 0{,}055\ \text{m} = 55\ \text{mm} \tag{4.11a}$$

For stage 1. Wheel tip diameter uses an addendum equal to the module. [20]

$d_{a,2\text{-}1}$ Stage 1: wheel tip diameter, mm.
$d_{2\text{-}1}$ Stage 1: wheel diameter, m.
m_1 Stage 1: module, m.

$$d_{a,2\text{-}1} = d_{2\text{-}1} + 2 \cdot m_1 \tag{4.12}$$

$$d_{a,2\text{-}1} = 0{,}275 + 2 \cdot 0{,}0025 = 0{,}28\ \text{m} = 280\ \text{mm} \tag{4.12a}$$

For stage 1. The pinion root diameter uses a dedendum of 1.25 modules. [20]

$d_{f,1}$ Stage 1: pinion root diameter, mm.
d_1 Stage 1: pinion diameter, m.
m_1 Stage 1: module, m.

$$d_{f,1} = d_1 - 2.5 \cdot m_1 \tag{4.13}$$

$$d_{f,1} = 0{,}05 - 2{,}5 \cdot 0{,}0025 = 0{,}04375\ \text{m} = 43{,}75\ \text{mm} \tag{4.13a}$$

For stage 1. The gear root diameter uses a dedendum of 1.25 modules. [20]

$d_{f,2\text{-}1}$ Stage 1: wheel root diameter, mm.
$d_{2\text{-}1}$ Stage 1: wheel diameter, m.
m_1 Stage 1: module, m.

$$d_{f,2\text{-}1} = d_{2\text{-}1} - 2.5 \cdot m_1 \tag{4.14}$$

$$d_{f,2\text{-}1} = 0{,}275 - 2{,}5 \cdot 0{,}0025 = 0{,}26875\ \text{m} = 268{,}75\ \text{mm} \tag{4.14a}$$

For stage 2. Transverse pressure angle follows from the normal pressure angle and helix angle β. [20]

α_{t2} Stage 2: transverse pressure angle, °.
α tooth pressure angle; adopted 20°, rad.
β_2 Stage 2: tooth helix angle β, dimensionless.

$$\alpha_{t2} = \tan^{-1}\left(\frac{\tan(\alpha)}{\cos(\beta_2)}\right) \tag{4.15}$$

$$\alpha_{t2} = \tan^{-1}\left(\frac{\tan(0{,}349066)}{\cos(0)}\right) = 0{,}34907 = 20\ ^\circ \tag{4.15a}$$

For stage 2. The worm pitch lead angle follows from its number of starts and diameter factor in the normal-module system. [20]

γ_2 Stage 2: worm lead angle, °.
q_{w2} Stage 2: worm diameter factor q, dimensionless.
$z_{\text{w},2}$ Stage 2: number of worm starts, dimensionless.

$$\gamma_2 = \sin^{-1}\left(\frac{z_{\text{w},2}}{q_{w2}}\right) \tag{4.16}$$

$$\gamma_2 = \sin^{-1}\left(\frac{2}{12}\right) = 0{,}16745 = 9{,}5941\ ^\circ \tag{4.16a}$$

For stage 2. For a simple planetary gear set, the ring tooth count equals the sun tooth count plus twice the planet tooth count. [20]

$z_{r,2}$ Stage 2: internal ring tooth count, dimensionless.

$z_{s,2}$ Stage 2: pinion tooth count, dimensionless.

$z_{p,2}$ Stage 2: planet tooth count, dimensionless.

$$z_{r,2} = z_{s,2} + 2 \cdot z_{p,2} \tag{4.17}$$

$$z_{r,2} = 24 + 2 \cdot 27 = 78 \tag{4.17a}$$

For stage 2. The pinion pitch cone angle follows from ratio and specified shaft angle for an external bevel pair. [20]

$\delta_{2\text{-}2}$ Stage 2: pinion pitch cone angle, °.

Σ_2 Stage 2: shaft angle Σ, dimensionless.

$z_{2\text{-}2}$ Stage 2: pinion tooth count, dimensionless.

z_3 Stage 2: wheel tooth count, dimensionless.

$$\delta_{2\text{-}2} = \cos^{-1}\left(\frac{\left(\frac{z_3}{z_{2\text{-}2}} + \cos\left(\Sigma_2\right)\right)}{\sqrt{\left(\frac{z_3}{z_{2\text{-}2}}\right)^2 + \frac{2 \cdot z_3}{z_{2\text{-}2}} \cdot \cos\left(\Sigma_2\right) + 1}}\right) \tag{4.18}$$

$$\delta_{2\text{-}2} = \cos^{-1}\left(\frac{\left(\frac{69}{20} + \cos\left(1{,}5708\right)\right)}{\sqrt{\left(\frac{69}{20}\right)^2 + \frac{2 \cdot 69}{20} \cdot \cos\left(1{,}5708\right) + 1}}\right) = 0{,}28212 = 16{,}164\ ^\circ \tag{4.18a}$$

For stage 2. The gear pitch cone angle equals the specified shaft angle minus the pinion cone angle. [20]

δ_3 Stage 2: wheel pitch cone angle, °.

Σ_2 Stage 2: shaft angle Σ, dimensionless.

$\delta_{2\text{-}2}$ Stage 2: pinion pitch cone angle, dimensionless.

$$\delta_3 = \Sigma_2 - \delta_{2\text{-}2} \tag{4.19}$$

$$\delta_3 = 1{,}5708 - 0{,}282124 = 1{,}2887 = 73{,}836\ ^\circ \tag{4.19a}$$

For stage 2. Outer cone distance follows from the pinion pitch diameter and cone angle. Face width must be smaller than this distance. [20]

R_2 Stage 2: outer cone distance, mm.
$\delta_{2\text{-}2}$ Stage 2: pinion pitch cone angle, dimensionless.
m_2 Stage 2: module, m.
$z_{2\text{-}2}$ Stage 2: pinion tooth count, dimensionless.

$$R_2 = \frac{m_2 \cdot z_{2\text{-}2}}{(2 \cdot \sin(\delta_{2\text{-}2}))} \tag{4.20}$$

$$R_2 = \frac{0{,}004 \cdot 20}{(2 \cdot \sin(0{,}282124))} = 0{,}14368\ \text{m} = 143{,}68\ \text{mm} \tag{4.20a}$$

For stage 2. The planet pitch diameter follows from the module and integer tooth count. [20]

d_{p2} Stage 2: planet pitch diameter, mm.
m_2 Stage 2: module, m.
$z_{\text{p},2}$ Stage 2: planet tooth count, dimensionless.

$$d_{p2} = m_2 \cdot z_{\text{p},2} \tag{4.21}$$

$$d_{p2} = 0{,}004 \cdot 27 = 0{,}108\ \text{m} = 108\ \text{mm} \tag{4.21a}$$

For stage 2. The pinion pitch diameter follows from the normal module, tooth count and helix angle β: the transverse module is $m_n/\cos\beta$. [20]

$d_{2\text{-}2}$ Stage 2: pinion diameter, mm.
β_2 Stage 2: tooth helix angle β, dimensionless.
m_2 Stage 2: module, m.
$z_{2\text{-}2}$ Stage 2: pinion tooth count, dimensionless.

$$d_{2\text{-}2} = \frac{m_2 \cdot z_{2\text{-}2}}{\cos(\beta_2)} \tag{4.22}$$

$$d_{2\text{-}2} = \frac{0{,}004 \cdot 20}{\cos(0)} = 0{,}08\ \text{m} = 80\ \text{mm} \tag{4.22a}$$

For stage 2. The gear pitch diameter uses the same normal module and helix angle β. [20]

d_3 Stage 2: wheel diameter, mm.
β_2 Stage 2: tooth helix angle β, dimensionless.
m_2 Stage 2: module, m.
z_3 Stage 2: wheel tooth count, dimensionless.

$$d_3 = \frac{m_2 \cdot z_3}{\cos(\beta_2)} \tag{4.23}$$

$$d_3 = \frac{0{,}004 \cdot 69}{\cos(0)} = 0{,}276\ \text{m} = 276\ \text{mm} \tag{4.23a}$$

For stage 2. Centre distance depends on the adopted mesh type. Bevel gear axes intersect, so this distance is zero; cone distance is calculated separately. [20]

a_{w2} Stage 2: centre distance, mm.
$d_{2\text{-}2}$ Stage 2: pinion diameter, m.
d_3 Stage 2: wheel diameter, m.

$$a_{w2} = \frac{(d_{2\text{-}2} + d_3)}{2} \tag{4.24}$$

$$a_{w2} = \frac{(0{,}08 + 0{,}276)}{2} = 0{,}178\ \text{m} = 178\ \text{mm} \tag{4.24a}$$

For stage 2. Pinion tip diameter uses an addendum equal to the module. [20]

$d_{a,2\text{-}2}$ Stage 2: pinion tip diameter, mm.
$d_{2\text{-}2}$ Stage 2: pinion diameter, m.
m_2 Stage 2: module, m.

$$d_{a,2\text{-}2} = d_{2\text{-}2} + 2 \cdot m_2 \tag{4.25}$$

$$d_{a,2\text{-}2} = 0{,}08 + 2 \cdot 0{,}004 = 0{,}088\ \text{m} = 88\ \text{mm} \tag{4.25a}$$

For stage 2. Wheel tip diameter uses an addendum equal to the module. [20]

$d_{a,3}$ Stage 2: wheel tip diameter, mm.
d_3 Stage 2: wheel diameter, m.
m_2 Stage 2: module, m.

$$d_{a,3} = d_3 + 2 \cdot m_2 \tag{4.26}$$

$$d_{a,3} = 0{,}276 + 2 \cdot 0{,}004 = 0{,}284\ \text{m} = 284\ \text{mm} \tag{4.26a}$$

For stage 2. The pinion root diameter uses a dedendum of 1.25 modules. [20]

$d_{f,2\text{-}2}$ Stage 2: pinion root diameter, mm.
$d_{2\text{-}2}$ Stage 2: pinion diameter, m.
m_2 Stage 2: module, m.

$$d_{f,2\text{-}2} = d_{2\text{-}2} - 2.5 \cdot m_2 \tag{4.27}$$

$$d_{f,2\text{-}2} = 0{,}08 - 2{,}5 \cdot 0{,}004 = 0{,}07\ \text{m} = 70\ \text{mm} \tag{4.27a}$$

For stage 2. The gear root diameter uses a dedendum of 1.25 modules. [20]

$d_{f,3}$ Stage 2: wheel root diameter, mm.
d_3 Stage 2: wheel diameter, m.
m_2 Stage 2: module, m.

$$d_{f,3} = d_3 - 2.5 \cdot m_2 \tag{4.28}$$

$$d_{f,3} = 0{,}276 - 2{,}5 \cdot 0{,}004 = 0{,}266\ \text{m} = 266\ \text{mm} \tag{4.28a}$$

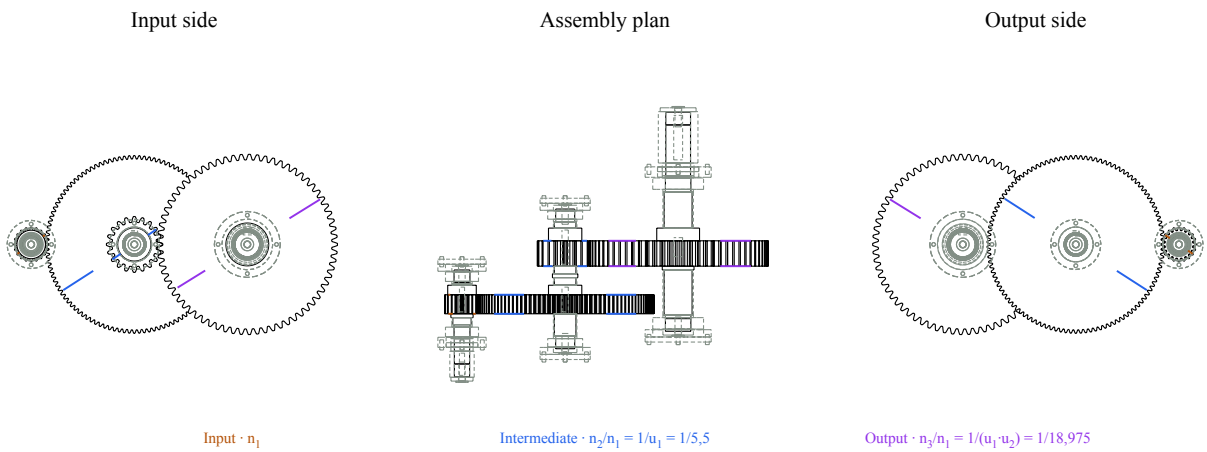


Figure 4.1 — Assembled cylindrical gearbox: plan and two end views at a common scale. Hidden lines removed. Root transition is approximate.

5. Tooth bending and contact

The specified factor K is applied once. Tooth allowable stresses and form factors depend on material, surface condition and life; they cannot be equated to steel yield strength. The Hertz estimate describes local elastic contact at the pitch point. Full contact and bending fatigue assessments require additional ISO 6336 factors.

Table 5.1 — Parameters and adoption

Parameter	Teaching value / alternatives
Stage 1: nameplate / catalogue torque rating $T_{rating1}$	0 N·m 0 — no rating specified. Adopt the rating of an actual product for the material, speed, life and service conditions; teaching geometry alone does not determine load capacity.
Stage 2: nameplate / catalogue torque rating $T_{rating2}$	0 N·m 0 — no rating specified. Adopt the rating of an actual product for the material, speed, life and service conditions; teaching geometry alone does not determine load capacity.
Stage 1: pinion tooth form factor Y_1	3
Stage 1: wheel tooth form factor $Y_{2\text{-}1}$	2,5
Stage 1: allowable bending stress SF_1	160 MPa
Stage 1: allowable local pressure SH_1	800 MPa

Parameter	Teaching value / alternatives
Stage 2: pinion tooth form factor $Y_{2\text{-}2}$	3
Stage 2: wheel tooth form factor Y_3	2,5
Stage 2: allowable bending stress SF_2	160 MPa
Stage 2: allowable local pressure SH_2	800 MPa

For stage 1. Axial force follows from the adopted mesh geometry. It is zero for a spur cylindrical stage; the worm relation excludes friction forces. [20]

F_{a1} Stage 1: output member axial force, N.
F_{t1} Stage 1: tangential force, N.
β_1 Stage 1: tooth helix angle β, dimensionless.

$$F_{a1} = F_{t1} \cdot \tan(\beta_1) \tag{5.1}$$

$$F_{a1} = 577{,}732 \cdot \tan(0) = 0\ \text{N} \tag{5.1a}$$

For stage 1. Tangential force follows from torque and working diameter of the adopted mesh; a planetary stage includes the planet count and uniform load sharing. [2]

F_{t1} Stage 1: tangential force, N.
T_1 Gearbox input torque, N·m.
d_1 Stage 1: pinion diameter, m.

$$F_{t1} = \frac{2 \cdot T_1}{d_1} \tag{5.2}$$

$$F_{t1} = \frac{2 \cdot 14{,}4433}{0{,}05} = 577{,}73\ \text{N} \tag{5.2a}$$

For stage 1. Radial force follows from tangential force and the adopted mesh angles: helix, cone or lead angle. [2]

F_{r1} Stage 1: radial force, N.
F_{t1} Stage 1: tangential force, N.

α tooth pressure angle; adopted 20°, rad.
β_1 Stage 1: tooth helix angle β, dimensionless.

$$F_{r1} = \frac{F_{t1} \cdot \tan(\alpha)}{\cos(\beta_1)} \tag{5.3}$$

$$F_{r1} = \frac{577{,}732 \cdot \tan(0{,}349066)}{\cos(0)} = 210{,}28\ \text{N} \tag{5.3a}$$

For stage 1. Pinion tooth bending stress uses the entered form factor and face width. [2]

$\sigma_{F,1}$ Stage 1: pinion bending, MPa.
F_{t1} Stage 1: tangential force, N.
K Design load factor, dimensionless.
Y_1 Stage 1: pinion tooth form factor, dimensionless.
b_1 Stage 1: face width, m.
m_1 Stage 1: module, m.

$$\sigma_{F,1} = \frac{F_{t1} \cdot K \cdot Y_1}{(b_1 \cdot m_1)} \tag{5.4}$$

$$\sigma_{F,1} = \frac{577{,}732 \cdot 1{,}25 \cdot 3}{(0{,}03 \cdot 0{,}0025)} = 28887000\ \text{Pa} = 28{,}887\ \text{MPa} \tag{5.4a}$$

For stage 1. Gear tooth bending stress uses its entered form factor. [2]

$\sigma_{F,2\text{-}1}$ Stage 1: wheel bending, MPa.
F_{t1} Stage 1: tangential force, N.
K Design load factor, dimensionless.
$Y_{2\text{-}1}$ Stage 1: wheel tooth form factor, dimensionless.
b_1 Stage 1: face width, m.
m_1 Stage 1: module, m.

$$\sigma_{F,2\text{-}1} = \frac{F_{t1} \cdot K \cdot Y_{2\text{-}1}}{(b_1 \cdot m_1)} \tag{5.5}$$

$$\sigma_{F,2\text{-}1} = \frac{577{,}732 \cdot 1{,}25 \cdot 2{,}5}{(0{,}03 \cdot 0{,}0025)} = 24072000\ \text{Pa} = 24{,}072\ \text{MPa} \tag{5.5a}$$

For stage 1. The reduced radius of curvature is evaluated at the involute pitch point of contact. [2]

ρ_1 Stage 1: reduced curvature radius, mm.
α_{t1} Stage 1: transverse pressure angle, dimensionless.
d_1 Stage 1: pinion diameter, m.
$d_{2\text{-}1}$ Stage 1: wheel diameter, m.

$$\rho_1 = \frac{\sin(\alpha_{t1}) \cdot d_1 \cdot d_{2\text{-}1}}{(2 \cdot (d_1 + d_{2\text{-}1}))} \tag{5.6}$$

$$\rho_1 = \frac{\sin(0{,}349066) \cdot 0{,}05 \cdot 0{,}275}{(2 \cdot (0{,}05 + 0{,}275))} = 0{,}007235\ \text{m} = 7{,}235\ \text{mm} \tag{5.6a}$$

For stage 1. Local Hertz contact pressure is estimated for elastic steel surfaces. This estimate does not replace a full ISO 6336 calculation. [2]

p_{H1} Stage 1: local Hertz pressure, MPa.
E Steel Young's modulus, Pa.
F_{t1} Stage 1: tangential force, N.
K Design load factor, dimensionless.
α_{t1} Stage 1: transverse pressure angle, dimensionless.
b_1 Stage 1: face width, m.
ν Poisson's ratio, dimensionless.
π number π, dimensionless.
ρ_1 Stage 1: reduced curvature radius, m.

$$p_{H1} = \sqrt{\frac{F_{t1} \cdot K \cdot E}{(2 \cdot (1 - \nu^2) \cdot \pi \cdot b_1 \cdot \rho_1 \cdot \cos(\alpha_{t1}))}} \tag{5.7}$$

$$p_{H1} = \sqrt{\frac{577{,}732 \cdot 1{,}25 \cdot 2{,}1 \cdot 10^{+11}}{(2 \cdot (1 - 0{,}3^2) \cdot 3{,}14159 \cdot 0{,}03 \cdot 0{,}00723504 \cdot \cos(0{,}349066))}} = 360610000\ \text{Pa} = 360{,}61\ \text{MPa} \tag{5.7a}$$

For stage 2. Axial force follows from the adopted mesh geometry. It is zero for a spur cylindrical stage; the worm relation excludes friction forces. [20]

F_{a2} Stage 2: output member axial force, N.
F_{t2} Stage 2: tangential force, N.
β_2 Stage 2: tooth helix angle β, dimensionless.

$$F_{a2} = F_{t2} \cdot \tan(\beta_2) \tag{5.8}$$

$$F_{a2} = 1946{,}24 \cdot \tan(0) = 0 \text{ N} \tag{5.8a}$$

For stage 2. Tangential force follows from torque and working diameter of the adopted mesh; a planetary stage includes the planet count and uniform load sharing. [2]

F_{t2} Stage 2: tangential force, N.
T_2 Stage 1: output torque, N·m.
$d_{2\text{-}2}$ Stage 2: pinion diameter, m.

$$F_{t2} = \frac{2 \cdot T_2}{d_{2\text{-}2}} \tag{5.9}$$

$$F_{t2} = \frac{2 \cdot 77{,}8494}{0{,}08} = 1946{,}2 \text{ N} \tag{5.9a}$$

For stage 2. Radial force follows from tangential force and the adopted mesh angles: helix, cone or lead angle. [2]

F_{r2} Stage 2: radial force, N.
F_{t2} Stage 2: tangential force, N.
α tooth pressure angle; adopted 20°, rad.
β_2 Stage 2: tooth helix angle β, dimensionless.

$$F_{r2} = \frac{F_{t2} \cdot \tan(\alpha)}{\cos(\beta_2)} \tag{5.10}$$

$$F_{r2} = \frac{1946{,}24 \cdot \tan(0{,}349066)}{\cos(0)} = 708{,}37 \text{ N} \tag{5.10a}$$

For stage 2. Pinion tooth bending stress uses the entered form factor and face width. [2]

$\sigma_{F,2\text{-}2}$ Stage 2: pinion bending, MPa.
F_{t2} Stage 2: tangential force, N.
K Design load factor, dimensionless.
$Y_{2\text{-}2}$ Stage 2: pinion tooth form factor, dimensionless.
b_2 Stage 2: face width, m.
m_2 Stage 2: module, m.

$$\sigma_{F,2\text{-}2} = \frac{F_{t2} \cdot K \cdot Y_{2\text{-}2}}{(b_2 \cdot m_2)} \tag{5.11}$$

$$\sigma_{F,2\text{-}2} = \frac{1946{,}24 \cdot 1{,}25 \cdot 3}{(0{,}04 \cdot 0{,}004)} = 45615000\ \text{Pa} = 45{,}615\ \text{MPa} \tag{5.11a}$$

For stage 2. Gear tooth bending stress uses its entered form factor. [2]

$\sigma_{F,3}$ Stage 2: wheel bending, MPa.
F_{t2} Stage 2: tangential force, N.
K Design load factor, dimensionless.
Y_3 Stage 2: wheel tooth form factor, dimensionless.
b_2 Stage 2: face width, m.
m_2 Stage 2: module, m.

$$\sigma_{F,3} = \frac{F_{t2} \cdot K \cdot Y_3}{(b_2 \cdot m_2)} \tag{5.12}$$

$$\sigma_{F,3} = \frac{1946{,}24 \cdot 1{,}25 \cdot 2{,}5}{(0{,}04 \cdot 0{,}004)} = 38012000\ \text{Pa} = 38{,}012\ \text{MPa} \tag{5.12a}$$

For stage 2. The reduced radius of curvature is evaluated at the involute pitch point of contact. [2]

ρ_2 Stage 2: reduced curvature radius, mm.
α_{t2} Stage 2: transverse pressure angle, dimensionless.
$d_{2\text{-}2}$ Stage 2: pinion diameter, m.
d_3 Stage 2: wheel diameter, m.

$$\rho_2 = \frac{\sin(\alpha_{t2}) \cdot d_{2\text{-}2} \cdot d_3}{(2 \cdot (d_{2\text{-}2} + d_3))} \tag{5.13}$$

$$\rho_2 = \frac{\sin(0{,}349066) \cdot 0{,}08 \cdot 0{,}276}{(2 \cdot (0{,}08 + 0{,}276))} = 0{,}010606\ \text{m} = 10{,}606\ \text{mm} \qquad (5.13a)$$

For stage 2. Local Hertz contact pressure is estimated for elastic steel surfaces. This estimate does not replace a full ISO 6336 calculation. [2]

p_{H2} Stage 2: local Hertz pressure, MPa.
E Steel Young's modulus, Pa.
F_{t2} Stage 2: tangential force, N.
K Design load factor, dimensionless.
α_{t2} Stage 2: transverse pressure angle, dimensionless.
b_2 Stage 2: face width, m.
ν Poisson's ratio, dimensionless.
π number π, dimensionless.
ρ_2 Stage 2: reduced curvature radius, m.

$$p_{H2} = \sqrt{\frac{F_{t2} \cdot K \cdot E}{(2 \cdot (1 - \nu^2) \cdot \pi \cdot b_2 \cdot \rho_2 \cdot \cos(\alpha_{t2}))}} \qquad (5.14)$$

$$p_{H2} = \sqrt{\frac{1946{,}24 \cdot 1{,}25 \cdot 2{,}1 \cdot 10^{+11}}{(2 \cdot (1 - 0{,}3^2) \cdot 3{,}14159 \cdot 0{,}04 \cdot 0{,}0106065 \cdot \cos(0{,}349066))}} = 473420000\ \text{Pa} = 473{,}42\ \text{MPa} \qquad (5.14a)$$

6. Low-speed shaft

Place the gear between bearings A and B. The tangential and radial load resultant defines the diagram plane. Before the gear Q = R_A, after it Q = $-R_B$; the Q jump is −F. Bending moment is continuous and zero at the bearings; its slope equals Q. Select material by grade, condition and original stock diameter. Adopt minimum static safety factor n and check $\sigma_{eq} \le R_e/[n]$. Also check deflection and twist. Fatigue and stress concentrations need separate assessment.
Table 6.1 — Parameters and adoption

Parameter	Teaching value / alternatives
Steel Young's modulus E	210 GPa
Poisson's ratio ν	0,3

Parameter	Teaching value / alternatives
External working-machine radial load $F_{external}$	0 N Additional force in the bending diagram plane. For a planetary carrier output, enter a nonzero external load; forces from evenly spaced planets are not added as one shaft force.
Bearing support span L_s	220 mm
Gear position from bearing A a_s	110 mm
Adopted shaft diameter d_s	45 mm
Shaft stock material and condition $q_{material}$	Custom properties and allowable stress; C45 047A +N; stock >10…<50 mm; C45 5081 +QT; stock >20…<40 mm; C45 5081 +QT; stock >40…<100 mm; 42CrMo4 6082 +QT; stock >25…<40 mm; 42CrMo4 6082 +QT; stock >40…<100 mm; 42CrMo4 6082 +QT; stock >100…<160 mm
Initial shaft stock diameter d_{stock}	55 mm The material property range applies to the original round stock. The finished shaft cannot exceed the stock diameter.
Stock minimum yield strength R_e	650 MPa
Required shaft static safety factor n_s	3 Adopted minimum n. For catalogue material [σ] = R_e/n; this is a static check without automatic transfer to gear tooth fatigue.
Custom allowable static shaft stress S_s	100 MPa
Permissible deflection at the gear f_{lim}	0,2 mm
Permissible angle of twist θ_{lim}	0,5 °
Output gear keyway length l_{gkey}	30 mm Teaching construction dimension. The keyway is centred in the hub seat and must fit within its length.
Output gear keyway width b_{gkey}	14 mm

Parameter	Teaching value / alternatives
Shaft end chamfers, 45° c_{shaft}	1 mm
Input shaft construction diameter d_{input}	30 mm Pinion seat. The shared construction checks stepped sections, keyways, pulley / sprocket load and both bearings; results follow in section 6.
Intermediate shaft construction diameter d_{middle}	35 mm Gear and pinion seats. The shared model checks individual sections, bending in two planes, torsion, axial force and bearings; results follow below.
Minimum input shaft overall length l_{input}	160 mm Desired minimum size. The shared construction model enlarges it if required by the hub, shoulder, bearings or seals; actual length is shown on the drawing and in the BOM.
Minimum intermediate shaft overall length l_{middle}	220 mm Minimum envelope. Actual length includes both gears, bearings and retainers; see the drawing and BOM.
Shaft shoulder height $h_{shoulder}$	2,5 mm Integral collar radial height. Teaching dimension; stock and hub mounting are checked.
Construction collar width $l_{shoulder}$	6 mm
Approach length with seal sleeve l_{seal}	20 mm Added before the coupling seat; the seal runs on a removable Ra 0.4 sleeve. Seal dimensions are teaching assumptions until a product is selected.
End retaining plate thickness t_{plate}	6 mm
Intermediate shaft second mesh-plane angle phi_{mesh}	0 ° Second-stage radial force direction relative to the first. Axial force sign depends on rotation/helix hand; a separate static design case is shown.

Multiply the final-stage tangential force by the load factor for low-speed shaft design. [2]

$F_{t,out}$ Shaft design tangential force, N.

F_{t2} Stage 2: tangential force, N.

K Design load factor, dimensionless.

$$F_{t,out} = K \cdot F_{t2} \tag{6.1}$$

$$F_{t,out} = 1{,}25 \cdot 1946{,}24 = 2432{,}8\ \text{N} \tag{6.1a}$$

The final mesh radial component includes its adopted type and load factor. [2]

$F_{r,out}$ Design radial force, N.
F_{r2} Stage 2: radial force, N.
K Design load factor, dimensionless.

$$F_{r,out} = K \cdot F_{r2} \tag{6.2}$$

$$F_{r,out} = 1{,}25 \cdot 708{,}372 = 885{,}47\ \text{N} \tag{6.2a}$$

The final mesh axial component includes the adopted type and load factor. It affects equivalent bearing loads. [20]

$F_{a,out}$ Output shaft design axial force, N.
F_{a2} Stage 2: output member axial force, N.
K Design load factor, dimensionless.

$$F_{a,out} = K \cdot F_{a2} \tag{6.3}$$

$$F_{a,out} = 1{,}25 \cdot 0 = 0\ \text{N} \tag{6.3a}$$

Shaft transverse load combines the resultant mesh force and specified external radial load. In a symmetric planetary carrier, planet forces balance; specify bearing load separately. [2]

F_{out} Resultant force at the gear, N.
$F_{external}$ External working-machine radial load, N.
$F_{r,out}$ Design radial force, N.
$F_{t,out}$ Shaft design tangential force, N.

$$F_{out} = \sqrt{{F_{t,out}}^2 + {F_{r,out}}^2} + F_{external} \tag{6.4}$$

$$F_{out} = \sqrt{2432{,}8^2 + 885{,}465^2} + 0 = 2588{,}9\ \text{N} \tag{6.4a}$$

Bearing A reaction follows from equilibrium of a simply supported shaft beam. [4]

R_A Bearing A radial reaction, N.
F_{out} Resultant force at the gear, N.
L_s Bearing support span, m.
a_s Gear position from bearing A, m.

$$R_A = \frac{F_{out} \cdot (L_s - a_s)}{L_s} \tag{6.5}$$

$$R_A = \frac{2588{,}93 \cdot (0{,}22 - 0{,}11)}{0{,}22} = 1294{,}5\ \text{N} \tag{6.5a}$$

Bearing B reaction follows from moment equilibrium about bearing A. [4]

R_B Bearing B radial reaction, N.
F_{out} Resultant force at the gear, N.
L_s Bearing support span, m.
a_s Gear position from bearing A, m.

$$R_B = \frac{F_{out} \cdot a_s}{L_s} \tag{6.6}$$

$$R_B = \frac{2588{,}93 \cdot 0{,}11}{0{,}22} = 1294{,}5\ \text{N} \tag{6.6a}$$

Between bearing A and the gear, shear force is constant and equals bearing A reaction. [4]

Q_{left} Shear force before the gear section, N.
R_A Bearing A radial reaction, N.

$$Q_{left} = R_A \tag{6.7}$$

$$Q_{left} = 1294{,}5\ \text{N} \tag{6.7a}$$

At the gear force, shear force jumps by that force and becomes the negative bearing B reaction. [4]

Q_{right} Shear force after the gear section, N.
R_B Bearing B radial reaction, N.

$$Q_{right} = -R_B \tag{6.8}$$

$$Q_{right} = -1294{,}5\ \text{N} \tag{6.8a}$$

For a point load between bearings, maximum bending moment occurs at the gear section. [4]

M_{max} Maximum bending moment, N·m.
F_{out} Resultant force at the gear, N.
L_s Bearing support span, m.
a_s Gear position from bearing A, m.

$$M_{max} = \frac{F_{out} \cdot a_s \cdot (L_s - a_s)}{L_s} \tag{6.9}$$

$$M_{max} = \frac{2588{,}93 \cdot 0{,}11 \cdot (0{,}22 - 0{,}11)}{0{,}22} = 142{,}39\ \text{N·m} \tag{6.9a}$$

Shaft design torque includes the load factor once. [4]

T_d Shaft design torque, N·m.
K Design load factor, dimensionless.
T_3 Actual output torque, N·m.

$$T_d = K \cdot T_3 \tag{6.10}$$

$$T_d = 1{,}25 \cdot 263{,}209 = 329{,}01\ \text{N·m} \tag{6.10a}$$

Normal bending stress uses a solid circular shaft cross-section. [4]

$\sigma_{b,s}$ Normal bending stress, MPa.
M_{max} Maximum bending moment, N·m.
d_s Adopted shaft diameter, m.
π number π, dimensionless.

$$\sigma_{b,s} = \frac{32 \cdot M_{max}}{\left(\pi \cdot {d_s}^3\right)} \tag{6.11}$$

$$\sigma_{b,s} = \frac{32 \cdot 142{,}391}{\left(3{,}14159 \cdot 0{,}045^3\right)} = 15916000 \text{ Pa} = 15{,}916 \text{ MPa} \tag{6.11a}$$

Torsional shear stress uses a solid circular shaft section. [4]

τ_s Torsional stress, MPa.
T_d Shaft design torque, N·m.
d_s Adopted shaft diameter, m.
π number π, dimensionless.

$$\tau_s = \frac{16 \cdot T_d}{\left(\pi \cdot {d_s}^3\right)} \tag{6.12}$$

$$\tau_s = \frac{16 \cdot 329{,}011}{\left(3{,}14159 \cdot 0{,}045^3\right)} = 18388000 \text{ Pa} = 18{,}388 \text{ MPa} \tag{6.12a}$$

Equivalent static stress uses the distortion-energy criterion for combined bending and torsion. [4]

$\sigma_{eq,s}$ Equivalent static stress, MPa.
$\sigma_{b,s}$ Normal bending stress, Pa.
τ_s Torsional stress, Pa.

$$\sigma_{eq,s} = \sqrt{{\sigma_{b,s}}^2 + 3 \cdot {\tau_s}^2} \tag{6.13}$$

$$\sigma_{eq,s} = \sqrt{1{,}59164 \cdot 10^{+7^2} + 3 \cdot 1{,}83884 \cdot 10^{+7^2}} = 35605000 \text{ Pa} = 35{,}605 \text{ MPa} \tag{6.13a}$$

For the selected material condition, static allowable stress follows from the minimum yield strength and adopted safety factor. Custom properties use the entered allowable stress. [22]

$[\sigma_s]$ Static allowable stress from the selected basis, MPa.
R_e Stock minimum yield strength, Pa.
n_s Required shaft static safety factor, dimensionless.

$$[\sigma_s] = \frac{R_e}{n_s} \tag{6.14}$$

$$[\sigma_s] = \frac{6{,}5 \cdot 10^{+8}}{3} = 216670000\ \text{Pa} = 216{,}67\ \text{MPa} \tag{6.14a}$$

Actual static safety factor is yield strength divided by equivalent stress. Compare with adopted minimum n; no artificial upper limit is imposed. [22]

n Actual safety factor against yielding, dimensionless.
R_e Stock minimum yield strength, Pa.
$\sigma_{eq,s}$ Equivalent static stress, Pa.

$$n = \frac{R_e}{\sigma_{eq,s}} \tag{6.15}$$

$$n = \frac{6{,}5 \cdot 10^{+8}}{3{,}56052 \cdot 10^{+7}} = 18{,}256 \tag{6.15a}$$

Estimate minimum diameter from static strength. The adopted construction diameter must equal or exceed it. [4]

d_{min} Diameter required by static strength, mm.
d_s Adopted shaft diameter, m.
$[\sigma_s]$ Static allowable stress from the selected basis, Pa.
$\sigma_{eq,s}$ Equivalent static stress, Pa.

$$d_{min} = d_s \cdot \left(\frac{\sigma_{eq,s}}{[\sigma_s]} \right)^{\left(\frac{1}{3}\right)} \tag{6.16}$$

$$d_{min} = 0{,}045 \cdot \left(\frac{3{,}56052 \cdot 10^{+7}}{2{,}16667 \cdot 10^{+8}} \right)^{\left(\frac{1}{3}\right)} = 0{,}024648 \text{ m} = 24{,}648 \text{ mm} \quad (6.16a)$$

Shear modulus follows from Young's modulus and Poisson's ratio for an isotropic elastic material. [2]

G Shear modulus, GPa.
E Steel Young's modulus, Pa.
ν Poisson's ratio, dimensionless.

$$G = \frac{E}{(2 \cdot (1 + \nu))} \quad (6.17)$$

$$G = \frac{2{,}1 \cdot 10^{+11}}{(2 \cdot (1 + 0{,}3))} = 80769000000 \text{ Pa} = 80{,}769 \text{ GPa} \quad (6.17a)$$

Calculate deflection at the force application point for constant EI. With asymmetric loading, this is not necessarily the maximum deflection. [2]

f_a Deflection at the gear, mm.
E Steel Young's modulus, Pa.
F_{out} Resultant force at the gear, N.
I Second moment of area, m^4.
L_s Bearing support span, m.
a_s Gear position from bearing A, m.

$$f_a = \frac{F_{out} \cdot {a_s}^2 \cdot (L_s - a_s)^2}{(3 \cdot E \cdot I \cdot L_s)} \quad (6.18)$$

$$f_a = \frac{2588{,}93 \cdot 0{,}11^2 \cdot (0{,}22 - 0{,}11)^2}{(3 \cdot 2{,}1 \cdot 10^{+11} \cdot 2{,}01289 \cdot 10^{-7} \cdot 0{,}22)} = 0{,}000013586 \text{ m} = 0{,}013586 \text{ mm} \quad (6.18a)$$

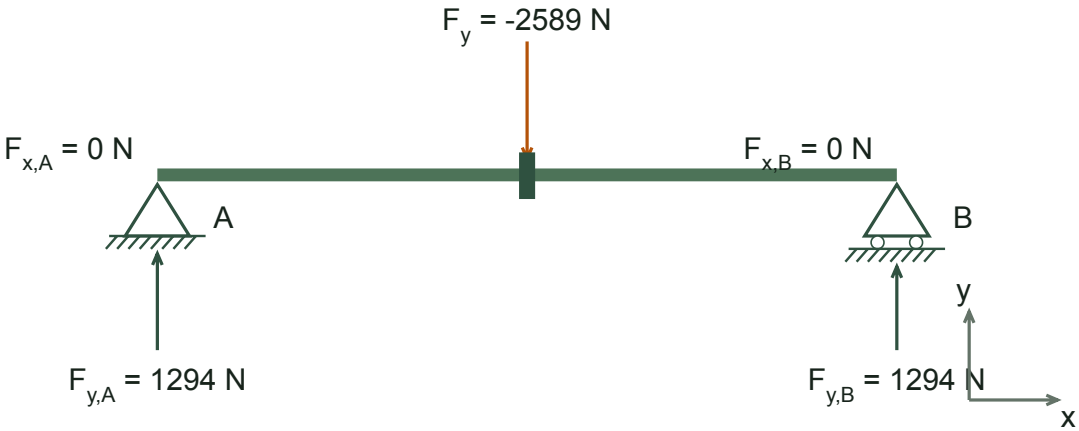
Twist between bearings uses constant torque and the polar second moment of area. [2]

θ Angle of twist between bearings, °.
G Shear modulus, Pa.
J Polar second moment of area, m^4.

L_s Bearing support span, m.
T_d Shaft design torque, N·m.

$$\theta = \frac{T_d \cdot L_s}{(G \cdot J)} \tag{6.19}$$

$$\theta = \frac{329{,}011 \cdot 0{,}22}{\left(8{,}07692 \cdot 10^{+10} \cdot 4{,}02578 \cdot 10^{-7}\right)} = 0{,}0022261 = 0{,}12754\ ^{\circ} \tag{6.19a}$$



L = 220 mm

a = 110 mm

T = 329.0 N·m

x: shaft axis; y: resultant force plane

Figure 6.1 — Low-speed shaft load diagram.

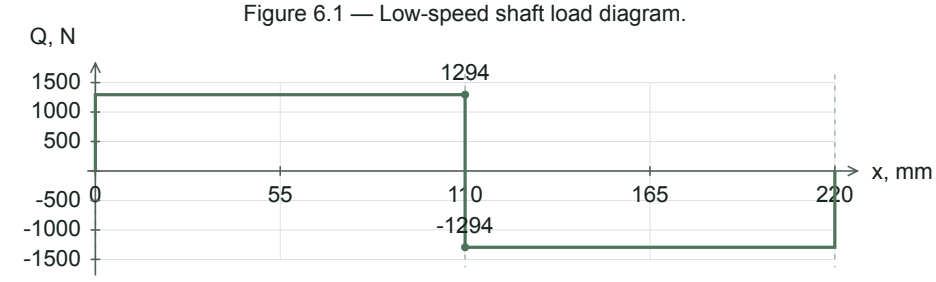


Figure 6.2 — Shear force

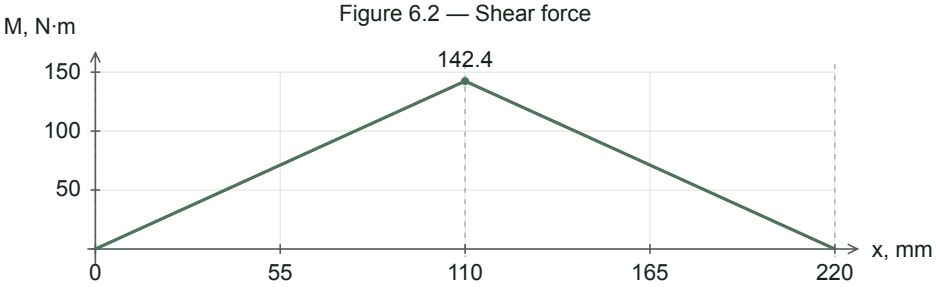


Figure 6.3 — Bending moment

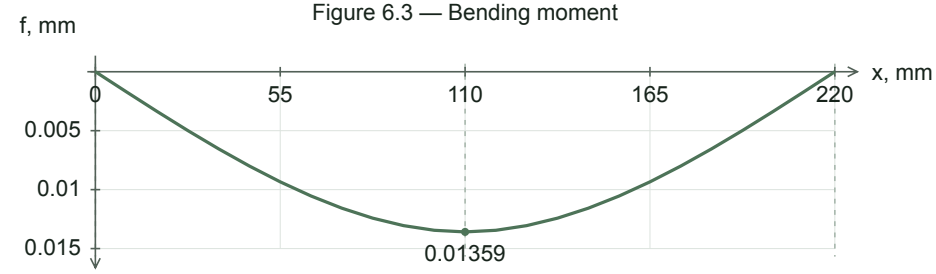


Figure 6.4 — Deflection

7. Rolling bearings

When opening an incompatible variant or exporting A1, compatible output shaft, hub, coupling and bearing dimensions are coordinated automatically. Both A1 buttons complete sizing and drawing in one click. If the journal is too small, “Select output shaft

and bearings” in section 7 prepares a separate dimensional proposal with keys and mounting approach; accept it after reviewing the changes table. Selection also checks actual stepped sections; other failed conditions are shown separately. Select an open SKF ball bearing from the shared CSV or enter data for a specific custom product. Saved C, C_0, bore and limiting speed must refer to one variant. Spur meshing has no axial force, so $P = R$. L_{10} life corresponds to 90% reliability. Check both bearings, nominal bore and limiting speed; adopt lubrication, clearances, fits and modified life separately.
Table 7.1 — Parameters and adoption

Parameter	**Teaching value / alternatives**
Bearing journal diameter d_j	40 mm Bearing journals may be smaller than the shaft centre. Torsion, axial force and stress concentration are checked; bearing bore must equal journal diameter. Bearing B mounts from the coupling end: the entire approach must be smaller than the journal by the adopted mounting diameter step.
Mounting diameter step before the journal $delta_{d,mount}$	1 mm 1 mm is an editable teaching design assumption, not a universal SKF/ISO rule. For example, use a ⌀39 h6 coupling approach before a ⌀40 k6 journal. Torsion and the key are checked for the smaller section; the journal seat diameter itself is unchanged.
Journal shoulder stress concentration factor K_j	1,5 1.5 is a teaching assumption for static assessment. Refine it using the diameter ratio and fillet radius; a separate fatigue calculation is required.
Bearing journal length l_j	25 mm Teaching construction dimension for the drawing; at least the selected bearing width. Adjust for the bearing location arrangement.
Shoulder fillet radius r_j	1 mm Teaching dimension. Check permissible bearing corner radius in the catalogue and stress concentration.

Parameter	Teaching value / alternatives
Bearing for A and B $q_{bearing}$	Custom bearing / manual catalogue data; SKF 6205, open; d = 25 mm; SKF 6206, open; d = 30 mm; SKF 6207, open; d = 35 mm; SKF 6208, open; d = 40 mm; SKF 6009, open; d = 45 mm; SKF 6209, open; d = 45 mm; SKF 6309, open; d = 45 mm; SKF 6210, open; d = 50 mm; SKF 6211, open; d = 55 mm; SKF 6212, open; d = 60 mm; SKF 6000, open; d = 10 mm; SKF 6200, open; d = 10 mm; SKF 6300, open; d = 10 mm; SKF 6001, open; d = 12 mm; SKF 6201, open; d = 12 mm; SKF 6301, open; d = 12 mm; SKF 6002, open; d = 15 mm; SKF 6202, open; d = 15 mm; SKF 6302, open; d = 15 mm; SKF 6003, open; d = 17 mm; SKF 6203, open; d = 17 mm; SKF 6303, open; d = 17 mm; SKF 6004, open; d = 20 mm; SKF 6204, open; d = 20 mm; SKF 6304, open; d = 20 mm; SKF 6005, open; d = 25 mm; SKF 6305, open; d = 25 mm; SKF 6006, open; d = 30 mm; SKF 6306, open; d = 30 mm; SKF 6007, open; d = 35 mm; SKF 6307, open; d = 35 mm; SKF 6008, open; d = 40 mm; SKF 6308, open; d = 40 mm; SKF 6010, open; d = 50 mm; SKF 6310, open; d = 50 mm; SKF 6011, open; d = 55 mm; SKF 6311, open; d = 55 mm; SKF 6012, open; d = 60 mm; SKF 6312, open; d = 60 mm; SKF 6013, open; d = 65 mm; SKF 6213, open; d = 65 mm; SKF 6313, open; d = 65 mm; SKF 6014, open; d = 70 mm; SKF 6214, open; d = 70 mm; SKF 6314, open; d = 70 mm; SKF 6015, open; d = 75 mm; SKF 6215, open; d = 75 mm; SKF 6315, open; d = 75 mm; SKF 6016, open; d = 80 mm; SKF 6216, open; d = 80 mm; SKF 6316, open; d = 80 mm; SKF 6017, open; d = 85 mm; SKF 6217, open; d = 85 mm; SKF 6317, open; d = 85 mm; SKF 6018, open; d = 90 mm; SKF 6218, open; d = 90 mm; SKF 6318, open; d = 90 mm; SKF 6019, open; d = 95 mm; SKF 6219, open; d = 95 mm; SKF 6319, open; d = 95 mm; SKF 6020, open; d = 100 mm; SKF 6220, open; d = 100 mm; SKF 618/5, open; d = 5 mm; SKF 619/5, open; d = 5 mm; SKF 625, open; d = 5 mm; SKF 635, open; d = 5 mm; SKF 618/6, open; d = 6 mm; SKF 619/6, open; d = 6 mm; SKF 626, open; d = 6 mm; SKF 618/7, open; d = 7 mm; SKF 619/7, open; d = 7 mm; SKF 607, open; d = 7 mm; SKF 627, open; d = 7 mm; SKF 618/8, open; d = 8 mm; SKF 619/8, open; d = 8 mm; SKF 608, open; d = 8 mm; SKF 628, open; d = 8 mm; SKF 618/9, open; d = 9 mm; SKF 619/9, open; d = 9 mm; SKF 609, open; d = 9 mm; SKF 629, open; d = 9 mm; SKF 61800, open; d = 10 mm; SKF 61900, open; d = 10 mm; SKF 16100, open; d = 10 mm; SKF 61801, open; d = 12 mm; SKF 61901, open; d = 12 mm; SKF 16101, open; d = 12 mm; SKF 61802, open; d = 15 mm; SKF 61902, open; d = 15 mm; SKF 16002, open; d = 15 mm; SKF 61803, open; d = 17 mm; SKF 61903, open; d = 17 mm; SKF 16003, open; d = 17 mm; SKF 6203 ETN9, open; d = 17 mm; SKF 6403, open; d = 17 mm; SKF 61804, open; d = 20 mm; SKF 61904, open; d = 20 mm; SKF 16004, open; d = 20 mm; SKF 6204 ETN9, open; d = 20 mm; SKF 6304 ETN9, open; d = 20 mm; SKF 6404, open; d = 20 mm; SKF 61805, open; d = 25 mm; SKF 61905, open; d = 25 mm; SKF 16005, open; d = 25 mm; SKF 6205 ETN9, open; d = 25 mm; SKF 6305 ETN9, open; d = 25 mm; SKF 6405, open; d = 25 mm; SKF 61806, open; d = 30 mm; SKF 61906, open; d = 30 mm; SKF 16006, open; d = 30 mm; SKF 6206 ETN9, open; d = 30 mm; SKF 6306 ETN9, open; d = 30 mm; SKF 6406, open; d = 30 mm; SKF 61807, open; d = 35 mm; SKF 61907, open; d = 35 mm; SKF

Parameter	Teaching value / alternatives
	16007, open; d = 35 mm; SKF 6207 ETN9, open; d = 35 mm; SKF 6307 M, open; d = 35 mm; SKF 6407, open; d = 35 mm; SKF 61808, open; d = 40 mm; SKF 61908, open; d = 40 mm; SKF 16008, open; d = 40 mm; SKF 6208 ETN9, open; d = 40 mm; SKF 6408, open; d = 40 mm; SKF 61809, open; d = 45 mm; SKF 61909, open; d = 45 mm; SKF 16009, open; d = 45 mm; SKF 6309 M, open; d = 45 mm; SKF 6409, open; d = 45 mm; SKF 61810, open; d = 50 mm; SKF 61910, open; d = 50 mm; SKF 16010, open; d = 50 mm; SKF 6210 M, open; d = 50 mm; SKF 6310 M, open; d = 50 mm; SKF 6410, open; d = 50 mm; SKF 61811, open; d = 55 mm; SKF 61911, open; d = 55 mm; SKF 16011, open; d = 55 mm; SKF 6011 M, open; d = 55 mm; SKF 6211 M, open; d = 55 mm; SKF 6311 M, open; d = 55 mm; SKF 6411, open; d = 55 mm; SKF 61812, open; d = 60 mm; SKF 61912, open; d = 60 mm; SKF 16012, open; d = 60 mm; SKF 6212 M, open; d = 60 mm; SKF 6312 M, open; d = 60 mm; SKF 6412, open; d = 60 mm; SKF 61813, open; d = 65 mm; SKF 61913, open; d = 65 mm; SKF 16013, open; d = 65 mm; SKF 6013 M, open; d = 65 mm; SKF 6213 M, open; d = 65 mm; SKF 6313 M, open; d = 65 mm; SKF 6413, open; d = 65 mm; SKF 61814, open; d = 70 mm; SKF 61914, open; d = 70 mm; SKF 16014, open; d = 70 mm; SKF 6014 M, open; d = 70 mm; SKF 6214 M, open; d = 70 mm; SKF 6314 M, open; d = 70 mm; SKF 6414, open; d = 70 mm; SKF 61815, open; d = 75 mm; SKF 61915, open; d = 75 mm; SKF 16015, open; d = 75 mm; SKF 6015 M, open; d = 75 mm; SKF 6215 M, open; d = 75 mm; SKF 6315 M, open; d = 75 mm; SKF 6415, open; d = 75 mm; SKF 61816, open; d = 80 mm; SKF 61916, open; d = 80 mm; SKF 16016, open; d = 80 mm; SKF 6216 M, open; d = 80 mm; SKF 6316 M, open; d = 80 mm; SKF 6416, open; d = 80 mm; SKF 61817, open; d = 85 mm; SKF 61917, open; d = 85 mm; SKF 16017, open; d = 85 mm; SKF 6217 M, open; d = 85 mm; SKF 6317 M, open; d = 85 mm; SKF 6417, open; d = 85 mm; SKF 61818, open; d = 90 mm; SKF 61918, open; d = 90 mm; SKF 16018, open; d = 90 mm; SKF 6018 M, open; d = 90 mm; SKF 6218 M, open; d = 90 mm; SKF 6318 M, open; d = 90 mm; SKF 6418, open; d = 90 mm; SKF 61819, open; d = 95 mm; SKF 61919, open; d = 95 mm; SKF 16019, open; d = 95 mm; SKF 6219 M, open; d = 95 mm; SKF 6319 M, open; d = 95 mm; SKF 61820, open; d = 100 mm; SKF 61920, open; d = 100 mm; SKF 16020, open; d = 100 mm; SKF 6020 M, open; d = 100 mm; SKF 6220 M, open; d = 100 mm; SKF 6320, open; d = 100 mm; SKF 6320 M, open; d = 100 mm; SKF 61821, open; d = 105 mm; SKF 61921, open; d = 105 mm; SKF 16021, open; d = 105 mm; SKF 6021, open; d = 105 mm; SKF 6021 M, open; d = 105 mm; SKF 6221, open; d = 105 mm; SKF 6321, open; d = 105 mm; SKF 61822, open; d = 110 mm; SKF 61922, open; d = 110 mm; SKF 61922 MA, open; d = 110 mm; SKF 16022, open; d = 110 mm; SKF 6022, open; d = 110 mm; SKF 6022 M, open; d = 110 mm; SKF 6222, open; d = 110 mm; SKF 6322, open; d = 110 mm; SKF 6322 M, open; d = 110 mm; SKF 61824, open; d = 120 mm; SKF 61924, open; d = 120 mm; SKF 61924 MA, open; d = 120 mm; SKF 16024, open; d = 120 mm; SKF 6024, open; d = 120 mm; SKF 6024 MA, open; d = 120 mm; SKF 6224, open; d = 120 mm; SKF 6224 M, open; d = 120 mm; SKF 6324, open; d = 120 mm; SKF 6324 M, open; d = 120 mm; SKF 61826, open; d = 130 mm;

Parameter	Teaching value / alternatives
	SKF 61926, open; d = 130 mm; SKF 16026, open; d = 130 mm; SKF 6026, open; d = 130 mm; SKF 6026 M, open; d = 130 mm; SKF 6226, open; d = 130 mm; SKF 6226 M, open; d = 130 mm; SKF 6326, open; d = 130 mm; SKF 6326 M, open; d = 130 mm; SKF 61828, open; d = 140 mm; SKF 61928, open; d = 140 mm; SKF 61928 MA, open; d = 140 mm; SKF 16028, open; d = 140 mm; SKF 6028, open; d = 140 mm; SKF 6028 M, open; d = 140 mm; SKF 6228, open; d = 140 mm; SKF 6228 MA, open; d = 140 mm; SKF 6328, open; d = 140 mm; SKF 6328 M, open; d = 140 mm; SKF 61830, open; d = 150 mm; SKF 61830 MA, open; d = 150 mm; SKF 61930 MA, open; d = 150 mm; SKF 16030, open; d = 150 mm; SKF 6030, open; d = 150 mm; SKF 6030 M, open; d = 150 mm; SKF 6230, open; d = 150 mm; SKF 6230 M, open; d = 150 mm; SKF 6330, open; d = 150 mm; SKF 6330 M, open; d = 150 mm; SKF 61832, open; d = 160 mm; SKF 61932, open; d = 160 mm; SKF 61932 MA, open; d = 160 mm; SKF 16032, open; d = 160 mm; SKF 6032, open; d = 160 mm; SKF 6032 M, open; d = 160 mm; SKF 6232, open; d = 160 mm; SKF 6232 M, open; d = 160 mm; SKF 6332, open; d = 160 mm; SKF 6332 M, open; d = 160 mm; SKF 61834, open; d = 170 mm; SKF 61934 MA, open; d = 170 mm; SKF 16034, open; d = 170 mm; SKF 6034, open; d = 170 mm; SKF 6034 M, open; d = 170 mm; SKF 6234, open; d = 170 mm; SKF 6234 M, open; d = 170 mm; SKF 6334, open; d = 170 mm; SKF 6334 M, open; d = 170 mm; SKF 61836, open; d = 180 mm; SKF 61936, open; d = 180 mm; SKF 61936 MA, open; d = 180 mm; SKF 16036, open; d = 180 mm; SKF 6036, open; d = 180 mm; SKF 6036 M, open; d = 180 mm; SKF 6236, open; d = 180 mm; SKF 6236 M, open; d = 180 mm; SKF 6336, open; d = 180 mm; SKF 6336 M, open; d = 180 mm; SKF 61838, open; d = 190 mm; SKF 61938, open; d = 190 mm; SKF 61938 MA, open; d = 190 mm; SKF 16038, open; d = 190 mm; SKF 6038, open; d = 190 mm; SKF 6038 M, open; d = 190 mm; SKF 6238, open; d = 190 mm; SKF 6238 M, open; d = 190 mm; SKF 6338, open; d = 190 mm; SKF 6338 M, open; d = 190 mm; SKF 61840, open; d = 200 mm; SKF 61940, open; d = 200 mm; SKF 61940 MA, open; d = 200 mm; SKF 16040, open; d = 200 mm; SKF 6040, open; d = 200 mm; SKF 6040 M, open; d = 200 mm; SKF 6240, open; d = 200 mm; SKF 6240 M, open; d = 200 mm; SKF 61844, open; d = 220 mm; SKF 61944, open; d = 220 mm; SKF 61944 MA, open; d = 220 mm; SKF 16044, open; d = 220 mm; SKF 6044, open; d = 220 mm; SKF 6044 M, open; d = 220 mm; SKF 6244, open; d = 220 mm; SKF 6244 M, open; d = 220 mm; SKF 6344 M, open; d = 220 mm; SKF 61848, open; d = 240 mm; SKF 61948, open; d = 240 mm; SKF 61948 MA, open; d = 240 mm; SKF 16048, open; d = 240 mm; SKF 16048 MA, open; d = 240 mm; SKF 6048, open; d = 240 mm; SKF 6048 M, open; d = 240 mm; SKF 6248 M, open; d = 240 mm; SKF 6348 M, open; d = 240 mm; SKF 61852, open; d = 260 mm; SKF 61952, open; d = 260 mm; SKF 61952 MA, open; d = 260 mm; SKF 16052, open; d = 260 mm; SKF 16052 MA, open; d = 260 mm; SKF 6052, open; d = 260 mm; SKF 6052 M, open; d = 260 mm; SKF 6252 M, open; d = 260 mm; SKF 61856, open; d = 280 mm; SKF 61856 MA, open; d = 280 mm; SKF 61956, open; d = 280 mm; SKF 61956 MA, open; d = 280 mm; SKF 16056,

Parameter	Teaching value / alternatives
	open; d = 280 mm; SKF 16056 MA, open; d = 280 mm; SKF 6056, open; d = 280 mm; SKF 6056 M, open; d = 280 mm; SKF 6256 M, open; d = 280 mm; SKF 61860, open; d = 300 mm; SKF 61860 MA, open; d = 300 mm; SKF 61960, open; d = 300 mm; SKF 61960 MA, open; d = 300 mm; SKF 16060 MA, open; d = 300 mm; SKF 6060 M, open; d = 300 mm; SKF 6260 M, open; d = 300 mm
Basic dynamic load rating C C	32,5 kN
Static load rating C_0 C_0	19 kN
Bearing bore diameter $d_{bearing}$	40 mm Must match the shaft journal diameter. For a custom product, 0 means unspecified size and an unchecked fit.
Bearing limiting speed $n_{bearing}$	11 000 min^{-1} Catalogue limiting speed for the open bearing; check lubrication and sealing conditions separately. For a custom product, 0 means unspecified speed.
Life exponent p_l	Ball bearing: p = 3; Roller bearing: p = 10/3
Equivalent radial load factor X $X_{bearing}$	0,56 Select X and Y for the specific bearing, Fa/C_0 ratio and clearance. Initial 0.56 / 1.6 values are an explicit teaching example; check the catalogue.
Equivalent axial load factor Y $Y_{bearing}$	1,6
Axial force share at bearing A $lambda_A$	1 1 — support A carries the axial force; 0 — support B. Sharing depends on the actual locating-bearing arrangement.
Required life H_{req}	10 000 h
Minimum static safety factor $s0_{req}$	1,5

Parameter	Teaching value / alternatives
Floating outer ring axial clearance g_{float}	1 mm

Check the bearing journal using design torque and axial force with the adopted stress concentration factor. Bending moment is zero at the ideal support centre; other sections and overhangs need additional checks. [4]

$\sigma_{eq,j}$ Journal equivalent stress, MPa.
$F_{a,out}$ Output shaft design axial force, N.
K_j Journal shoulder stress concentration factor, dimensionless.
T_d Shaft design torque, N·m.
d_j Bearing journal diameter, m.
π number π, dimensionless.

$$\sigma_{eq,j} = K_j \cdot \sqrt{\left(\frac{4 \cdot F_{a,out}}{\left(\pi \cdot {d_j}^2\right)}\right)^2 + 3 \cdot \left(\frac{16 \cdot T_d}{\left(\pi \cdot {d_j}^3\right)}\right)^2} \tag{7.1}$$

$$\sigma_{eq,j} = 1{,}5 \cdot \sqrt{\left(\frac{4 \cdot 0}{\left(3{,}14159 \cdot 0{,}04^2\right)}\right)^2 + 3 \cdot \left(\frac{16 \cdot 329{,}011}{\left(3{,}14159 \cdot 0{,}04^3\right)}\right)^2} = 68023000\ \text{Pa} = 68{,}023\ \text{MPa} \tag{7.1a}$$

The preliminary journal minimum uses torsion and allowable equivalent stress. A separate check also includes axial force. [4]

$d_{j,min}$ Minimum journal diameter from torsion, mm.
K_j Journal shoulder stress concentration factor, dimensionless.
T_d Shaft design torque, N·m.
π number π, dimensionless.
$[\sigma_s]$ Static allowable stress from the selected basis, Pa.

$$d_{j,min} = \left(\frac{16 \cdot \sqrt{3} \cdot K_j \cdot T_d}{\left(\pi \cdot [\sigma_s]\right)}\right)^{\left(\frac{1}{3}\right)} \tag{7.2}$$

$$d_{j,min} = \left(\frac{16 \cdot \sqrt{3} \cdot 1{,}5 \cdot 329{,}011}{(3{,}14159 \cdot 2{,}16667 \cdot 10^{+8})} \right)^{\left(\frac{1}{3}\right)} = 0{,}027186 \text{ m} = 27{,}186 \text{ mm} \quad (7.2a)$$

Bearing A equivalent dynamic load uses its radial reaction and adopted axial force share. Select X and Y from the catalogue for the specific bearing and load ratio. [23]

P_A Bearing A equivalent dynamic load, N.
$F_{a,out}$ Output shaft design axial force, N.
R_A Bearing A radial reaction, N.
$X_{bearing}$ Equivalent radial load factor X, dimensionless.
$Y_{bearing}$ Equivalent axial load factor Y, dimensionless.
$lambda_A$ Axial force share at bearing A, dimensionless.

$$P_A = \max\left(R_A, X_{bearing} \cdot R_A + Y_{bearing} \cdot F_{a,out} \cdot lambda_A\right) \quad (7.3)$$

$$P_A = \max\left(1294{,}46, 0{,}56 \cdot 1294{,}46 + 1{,}6 \cdot 0 \cdot 1\right) = 1294{,}5 \text{ N} \quad (7.3a)$$

Bearing B equivalent dynamic load uses its radial reaction and remaining axial force. The adopted axial load share depends on the locating and floating bearing arrangement. [23]

P_B Bearing B equivalent dynamic load, N.
$F_{a,out}$ Output shaft design axial force, N.
R_B Bearing B radial reaction, N.
$X_{bearing}$ Equivalent radial load factor X, dimensionless.
$Y_{bearing}$ Equivalent axial load factor Y, dimensionless.
$lambda_A$ Axial force share at bearing A, dimensionless.

$$P_B = \max\left(R_B, X_{bearing} \cdot R_B + Y_{bearing} \cdot F_{a,out} \cdot (1 - lambda_A)\right) \quad (7.4)$$

$$P_B = \max\left(1294{,}46, 0{,}56 \cdot 1294{,}46 + 1{,}6 \cdot 0 \cdot (1 - 1)\right) = 1294{,}5 \text{ N} \quad (7.4a)$$

Deep groove ball bearing A equivalent static load is the greater of the radial load and the sum of 0.6 times the radial load and 0.5 times the axial load. [23]

$P_{0,A}$ Bearing A equivalent static load, N.
$F_{a,out}$ Output shaft design axial force, N.
R_A Bearing A radial reaction, N.

$lambda_A$ Axial force share at bearing A, dimensionless.

$$P_{0,A} = \max\left(R_A,\ 0.6 \cdot R_A + 0.5 \cdot F_{a,out} \cdot lambda_A\right) \tag{7.5}$$

$$P_{0,A} = \max\left(1294{,}46,\ 0{,}6 \cdot 1294{,}46 + 0{,}5 \cdot 0 \cdot 1\right) = 1294{,}5\ \mathrm{N} \tag{7.5a}$$

Bearing B equivalent static load uses its radial and adopted axial forces. [23]

$P_{0,B}$ Bearing B equivalent static load, N.
$F_{a,out}$ Output shaft design axial force, N.
R_B Bearing B radial reaction, N.
$lambda_A$ Axial force share at bearing A, dimensionless.

$$P_{0,B} = \max\left(R_B,\ 0.6 \cdot R_B + 0.5 \cdot F_{a,out} \cdot (1 - lambda_A)\right) \tag{7.6}$$

$$P_{0,B} = \max\left(1294{,}46,\ 0{,}6 \cdot 1294{,}46 + 0{,}5 \cdot 0 \cdot (1 - 1)\right) = 1294{,}5\ \mathrm{N} \tag{7.6a}$$

Support A basic rating life is determined from equivalent dynamic load at 90% reliability. The SI result is in seconds; divide by 3600 for hours. [23]

$L_{10,A}$ Bearing A basic rating life, h.
C Basic dynamic load rating C, N.
P_A Bearing A equivalent dynamic load, N.
10^6 basic revolution count for rating life, dimensionless.
n_3 Actual output speed, s^{-1}.
p_l Life exponent, dimensionless.
q_{life} radial load multiplier; 1 in the current check, dimensionless.

$$L_{10,A} = \frac{10^6}{(n_3)} \cdot \left(\frac{C}{(P_A \cdot q_{life})}\right)^{p_l} \tag{7.7}$$

$$L_{10,A} = \frac{1 \cdot 10^{+6}}{(1{,}26482)} \cdot \left(\frac{32500}{(1294{,}46 \cdot 1)}\right)^{3} = 12513000000\ \mathrm{s} = 3475800\ \mathrm{h} \tag{7.7a}$$

Support B basic rating life is determined from its equivalent dynamic load and shaft speed; lubrication-adjusted life is not included. [23]

$L_{10,B}$ Bearing B basic rating life, h.
C Basic dynamic load rating C, N.
P_B Bearing B equivalent dynamic load, N.
10^6 basic revolution count for rating life, dimensionless.
n_3 Actual output speed, s^{-1}.
p_l Life exponent, dimensionless.
q_{life} radial load multiplier; 1 in the current check, dimensionless.

$$L_{10,B} = \frac{10^6}{(n_3)} \cdot \left(\frac{C}{(P_B \cdot q_{life})} \right)^{p_l} \tag{7.8}$$

$$L_{10,B} = \frac{1 \cdot 10^{+6}}{(1{,}26482)} \cdot \left(\frac{32500}{(1294{,}46 \cdot 1)} \right)^{3} = 12513000000 \text{ s} = 3475800 \text{ h} \tag{7.8a}$$

Bearing A static safety factor is static load rating divided by equivalent static load. [23]

$s_{0,A}$ Bearing A static safety factor, dimensionless.
C_0 Static load rating C_0, N.
$P_{0,A}$ Bearing A equivalent static load, N.

$$s_{0,A} = \frac{C_0}{P_{0,A}} \tag{7.9}$$

$$s_{0,A} = \frac{19000}{1294{,}46} = 14{,}678 \tag{7.9a}$$

Bearing B static safety factor is static load rating divided by equivalent static load. [23]

$s_{0,B}$ Bearing B static safety factor, dimensionless.
C_0 Static load rating C_0, N.
$P_{0,B}$ Bearing B equivalent static load, N.

$$s_{0,B} = \frac{C_0}{P_{0,B}} \tag{7.10}$$

$$s_{0,B} = \frac{19000}{1294{,}46} = 14{,}678 \tag{7.10a}$$

Shaft, bearing and hub design

First define assembly order and axial force paths, then specify shoulders and fits. Fit, torque transmission and axial retention are separate requirements.

Table 7.2 — Construction design rules

Element	What to determine
Mounting and dismounting	Each part must pass from a free end to its seat. Provide a smaller diameter before the seating journal and puller access. ⌀39 h6 before ⌀40 k6 is a teaching example. The editable 1 mm step is not a universal rule.
Datums and dimensions	The output shaft journal axes establish common datum A–B. For the gear, A is the bore axis and B is the hub abutment face. Dimension axial positions from a functional stop; check the dimension chain, seat runout and face perpendicularity.
Fits and surface texture	Select tolerance zones for ring rotation relative to the load, temperature and duty. Journal k6 and gear bore H7 are teaching assumptions. Clearance along the mounting path does not loosen the working fit. Specify Ra and geometric tolerances for each functional surface.
Shoulders, fillets and chamfers	Obtain the abutment diameter d_a and permissible radius r_a for the selected bearing. The fillet must clear the ring face; a larger radius requires a compatible spacer. An entry chamfer facilitates assembly. R1 and 1 × 45° are editable assumptions here, not catalogue limits.
Manufacturing relief grooves	Use a relief groove per ISO 18388 for tool or grinding-wheel clearance only where machining requires it. Do not confuse it with a retaining-ring groove. Grooves, threads and keyways reduce the section and require strength and fatigue checks.
Bearing axial location	Identify the locating and floating bearings. Restrain both rings of the locating bearing axially in both directions. Options include a shoulder and nut with a lock washer, an end plate or an approved retaining ring; a thread is needed only for the selected method. The floating arrangement must allow thermal expansion.

Element	What to determine
Gears, pulleys and couplings	The key transmits torque; define hub axial stops in both directions separately. Coordinate shoulder, spacer and nut or end plate with hub width, assembly direction and axial force. For a bevel pair, the abutment face position determines mounting distance and meshing. All rings, sleeves and fasteners must appear in the assembly and BOM.

Sources: NTN: mounting, fits and abutments; SKF: Maintenance handbook, pp. 37–38; KG: gear retention, pp. 51–53; ISO 5459:2024 — datums; ISO 18388:2016 — relief grooves.

The shared construction model defines collars, removable stepped sleeves, hubs, end plates, M10 axial bolts, covers and seals. Parts share identifiers across the assembly, BOM, drawings and assembly sequence. Mounting paths, axial positioning and individual shaft sections are checked. Covers and seal envelopes are teaching dimensions; tightening, locking, fatigue and final seal selection require product-specific checks.

8. Key and output coupling

Select key dimensions for the specific shaft diameter and hub. A round-ended key has effective length l − b. Check bearing stress and shear separately. Compare coupling design torque with the product rating for the adopted duty; do not increase the rating without selecting another product.

Table 8.1 — Parameters and adoption

Parameter	Teaching value / alternatives
Output coupling seat diameter d_{cseat}	39 mm The seat and entire bearing B mounting path must be smaller than the journal by the adopted mounting diameter step. Check the reduced section for torque, key bearing stress and shear. Match coupling bore to this seat.
Output coupling seat length l_{cseat}	80 mm
Coupling keyway offset from shoulder x_{kc}	5 mm
Shaft keyway depth t_{key}	5,5 mm
Key width b_k	14 mm
Key height h_k	9 mm
Key overall length l_k	70 mm

Parameter	Teaching value / alternatives
Allowable key bearing stress S_k	100 MPa
Allowable key shear stress τ_k	60 MPa
Output coupling torque rating T_{clim}	500 N·m

Check the output coupling seat using design torque, its own diameter and the adopted stress concentration factor. This static estimate does not replace keyway fatigue assessment. [4]

$\sigma_{eq,c}$ Coupling seat equivalent stress, MPa.
K_j Journal shoulder stress concentration factor, dimensionless.
T_d Shaft design torque, N·m.
d_{cseat} Output coupling seat diameter, m.
π number π, dimensionless.

$$\sigma_{eq,c} = \frac{K_j \cdot \sqrt{3} \cdot 16 \cdot T_d}{\left(\pi \cdot {d_{cseat}}^3\right)} \tag{8.1}$$

$$\sigma_{eq,c} = \frac{1{,}5 \cdot \sqrt{3} \cdot 16 \cdot 329{,}011}{\left(3{,}14159 \cdot 0{,}039^3\right)} = 73390000\ \text{Pa} = 73{,}39\ \text{MPa} \tag{8.1a}$$

For a round-ended key, effective length equals overall length minus width. [5]

l_e Key effective length, mm.
b_k Key width, m.
l_k Key overall length, m.

$$l_e = l_k - b_k \tag{8.2}$$

$$l_e = 0{,}07 - 0{,}014 = 0{,}056\ \text{m} = 56\ \text{mm} \tag{8.2a}$$

Key bearing stress uses working area $(h/2)l_e$ and design torque. [5]

$\sigma_{зм}$ Key bearing stress, MPa.
T_d Shaft design torque, N·m.

d_{cseat} Output coupling seat diameter, m.
h_k Key height, m.
l_e Key effective length, m.

$$\sigma_{зм} = \frac{4 \cdot T_d}{(d_{cseat} \cdot h_k \cdot l_e)} \tag{8.3}$$

$$\sigma_{зм} = \frac{4 \cdot 329{,}011}{(0{,}039 \cdot 0{,}009 \cdot 0{,}056)} = 66954000\ \text{Pa} = 66{,}954\ \text{MPa} \tag{8.3a}$$

Key shear stress uses working area b l_e. [5]

τ_k Key shear, MPa.
T_d Shaft design torque, N·m.
b_k Key width, m.
d_{cseat} Output coupling seat diameter, m.
l_e Key effective length, m.

$$\tau_k = \frac{2 \cdot T_d}{(d_{cseat} \cdot b_k \cdot l_e)} \tag{8.4}$$

$$\tau_k = \frac{2 \cdot 329{,}011}{(0{,}039 \cdot 0{,}014 \cdot 0{,}056)} = 21521000\ \text{Pa} = 21{,}521\ \text{MPa} \tag{8.4a}$$

The output coupling torque rating must equal or exceed the low-speed shaft design torque. [2]

$T_{coupling}$ Required output coupling torque, N·m.
T_d Shaft design torque, N·m.

$$T_{coupling} = T_d \tag{8.5}$$

$$T_{coupling} = 329{,}01\ \text{N·m} \tag{8.5a}$$

9. Bolted joint

Specify external load, bolt count, preload and load fraction χ according to joint stiffness. First check residual joint compression. After separation, results from the linear χ model do not establish serviceability. Friction resists slip; shank shear is a separate assessment after slip. Threads in the shear plane, bearing stress, eccentricity and tightening scatter require additional checks.
Table 9.1 — Parameters and adoption

Parameter	Teaching value / alternatives
Nominal bolt diameter d_{bolt}	12 mm
Metric thread pitch p_{bolt}	1,75 mm
Bolt count N_{bolt}	4
Preload per bolt F_{pre}	10 kN
External axial force F_{ax}	12 kN
Transverse force F_{sh}	4 kN
Axial load fraction χ χ	0,2
Joint friction coefficient μ	0,15
Allowable bolt stress S_{bolt}	300 MPa
Approximate tightening factor K_T	0,2

Estimate metric thread tensile stress area using nominal diameter and pitch. [37]

A_s Thread tensile stress area, mm².
d_{bolt} Nominal bolt diameter, m.
p_{bolt} Metric thread pitch, m.
π number π, dimensionless.

$$A_s = \frac{\pi \cdot (d_{bolt} - 0.9382 \cdot p_{bolt})^2}{4} \tag{9.1}$$

$$A_s = \frac{3{,}14159 \cdot (0{,}012 - 0{,}9382 \cdot 0{,}00175)^2}{4} = 0{,}000084266\ \text{m}^2 = 84{,}266\ \text{mm}^2 \tag{9.1a}$$

Axial force per bolt assumes a compressed joint and uniform load distribution between bolts. [3]

F_b Axial force per bolt, kN.

F_{ax} External axial force, N.
F_{pre} Preload per bolt, N.
N_{bolt} Bolt count, dimensionless.
χ Axial load fraction χ, dimensionless.

$$F_b = F_{pre} + \frac{\chi \cdot F_{ax}}{N_{bolt}} \tag{9.2}$$

$$F_b = 10000 + \frac{0{,}2 \cdot 12000}{4} = 10600\ \text{N} = 10{,}6\ \text{kN} \tag{9.2a}$$

Residual joint compression must be positive. After separation, axial load sharing by χ is invalid. [3]

F_{cl} Residual compression per bolt, kN.
F_{ax} External axial force, N.
F_{pre} Preload per bolt, N.
N_{bolt} Bolt count, dimensionless.
χ Axial load fraction χ, dimensionless.

$$F_{cl} = F_{pre} - \frac{(1-\chi) \cdot F_{ax}}{N_{bolt}} \tag{9.3}$$

$$F_{cl} = 10000 - \frac{(1-0{,}2) \cdot 12000}{4} = 7600\ \text{N} = 7{,}6\ \text{kN} \tag{9.3a}$$

Bolt tensile stress uses the thread tensile stress area while the joint remains compressed. [3]

$\sigma_{b,bolt}$ Bolt tensile stress, MPa.
A_s Thread tensile stress area, m².
F_b Axial force per bolt, N.

$$\sigma_{b,bolt} = \frac{F_b}{A_s} \tag{9.4}$$

$$\sigma_{b,bolt} = \frac{10600}{8{,}42664 \cdot 10^{-5}} = 125790000\ \text{Pa} = 125{,}79\ \text{MPa} \tag{9.4a}$$

Joint friction resistance uses residual compression from all bolts; a separated joint cannot transmit transverse force by friction. [3]

F_{fr} Joint friction resistance to slip, kN.
F_{cl} Residual compression per bolt, N.
N_{bolt} Bolt count, dimensionless.
μ Joint friction coefficient, dimensionless.
0 force zero, N.

$$F_{fr} = \mu \cdot N_{bolt} \cdot \max\left(F_{cl}, 0\right) \tag{9.5}$$

$$F_{fr} = 0{,}15 \cdot 4 \cdot \max\left(7600, 0\right) = 4560\ \text{N} = 4{,}56\ \text{kN} \tag{9.5a}$$

After slip, estimate plain shank shear in one plane with uniformly loaded bolts. [3]

τ_b Stress after slip: shank shear, MPa.
F_{sh} Transverse force, N.
N_{bolt} Bolt count, dimensionless.
d_{bolt} Nominal bolt diameter, m.
π number π, dimensionless.

$$\tau_b = \frac{4 \cdot F_{sh}}{\left(N_{bolt} \cdot \pi \cdot {d_{bolt}}^2\right)} \tag{9.6}$$

$$\tau_b = \frac{4 \cdot 4000}{\left(4 \cdot 3{,}14159 \cdot 0{,}012^2\right)} = 8841900\ \text{Pa} = 8{,}8419\ \text{MPa} \tag{9.6a}$$

Equivalent stress combines tension and shear in a plain bolt shank; threads in the shear plane are excluded. [3]

$\sigma_{eq,bolt}$ Combined tension and shank shear estimate, MPa.
$\sigma_{b,bolt}$ Bolt tensile stress, Pa.
τ_b Stress after slip: shank shear, Pa.

$$\sigma_{eq,bolt} = \sqrt{{\sigma_{b,bolt}}^2 + 3 \cdot {\tau_b}^2} \tag{9.7}$$

$$\sigma_{eq,bolt} = \sqrt{1{,}25792 \cdot 10^{+8^2} + 3 \cdot 8{,}84194 \cdot 10^{+6^2}} = 126720000\ \text{Pa} = 126{,}72\ \text{MPa} \tag{9.7a}$$

Estimate tightening torque using the adopted factor K_t. Account separately for friction and tool accuracy scatter. [38]

M_A Approximate tightening torque, N·m.
F_{pre} Preload per bolt, N.
K_T Approximate tightening factor, dimensionless.
d_{bolt} Nominal bolt diameter, m.

$$M_A = K_T \cdot F_{pre} \cdot d_{bolt} \tag{9.8}$$

$$M_A = 0{,}2 \cdot 10000 \cdot 0{,}012 = 24\ \text{N·m} \tag{9.8a}$$

10. Basic machine-element calculations

The rod and spring are separate examples, not drive loads. The rod uses linear elasticity. Adopt the physical material limit and required safety factor $[n] > 1$: $[\sigma] = R_e/[n]$, $n = R_e/\sigma \geq [n]$. The spring similarly uses its limiting shear stress. Do not apply a second safety factor to an already specified allowable stress. Teaching values [n] = 2 and 1.5 require justification for the duty; they are not universal requirements. Use $D/d \geq 4$ and spring-wire properties for the spring. Check stress with curvature correction, rate and working travel; coil contact, buckling and fatigue are not calculated here.

Table 10.1 — Parameters and adoption

Parameter	Teaching value / alternatives
Rod axial force F_{rod}	10 kN
Rod cross-sectional area A_{rod}	100 mm²
Rod length L_{rod}	500 mm
Strength limit input method q_{basic}	Ready-made allowable stresses (without a physical safety factor); Material strength limits and required safety factor

Parameter	Teaching value / alternatives
Adopted rod material yield strength R_{rod}	300 MPa 300 MPa is an adopted teaching value; use the verified yield strength for the actual material condition and size.
Required rod safety factor [n] n_{rod}	2 Adopted static tensile minimum; [σ] = R_e/[n]. The value 2 is a teaching choice, not a universal requirement for all materials.
Wire-material limiting shear stress R_{sp}	600 MPa 600 MPa is a teaching assumption. A verified wire-material limit and failure criterion are required; fatigue life is checked separately.
Required spring safety factor [n] n_{sp}	1,5 Adopted minimum based on limiting shear stress; [τ] = τ_lim/[n]. Fatigue requires separate data.
Allowable rod stress S_{rod}	160 MPa
Spring force F_{sp}	100 N
Spring wire diameter d_w	4 mm
Mean spring diameter D_{sp}	30 mm
Active coils N_{sp}	8
Allowable wire stress τ_{sp}	400 MPa
Spring travel / wheel travel ratio i_{motion}	0,8

Axial tensile stress uses a uniformly loaded rod of constant cross-section. [2]

σ_{rod} Axial tensile stress, MPa.
A_{rod} Rod cross-sectional area, m².
F_{rod} Rod axial force, N.

$$\sigma_{rod} = \frac{F_{rod}}{A_{rod}} \tag{10.1}$$

$$\sigma_{rod} = \frac{10000}{1 \cdot 10^{-4}} = 100000000\ \text{Pa} = 100\ \text{MPa} \tag{10.1a}$$

Use the entered allowable stress or calculate it from yield strength and the required static safety factor. [40]

$[\sigma]_{rod}$ Allowable rod stress, MPa.
R_{rod} Adopted rod material yield strength, Pa.
n_{rod} Required rod safety factor [n], dimensionless.

$$[\sigma]_{rod} = \frac{R_{rod}}{n_{rod}} \tag{10.2}$$

$$[\sigma]_{rod} = \frac{3 \cdot 10^{+8}}{2} = 150000000\ \text{Pa} = 150\ \text{MPa} \tag{10.2a}$$

Actual yield safety factor uses the verified or explicitly adopted material limit; compare with required [n] > 1. [40]

n_{rod} Actual rod safety factor, dimensionless.
R_{rod} Adopted rod material yield strength, Pa.
σ_{rod} Axial tensile stress, Pa.

$$n_{rod} = \frac{R_{rod}}{\sigma_{rod}} \tag{10.3}$$

$$n_{rod} = \frac{3 \cdot 10^{+8}}{1 \cdot 10^{+8}} = 3 \tag{10.3a}$$

Strain follows Hooke's law within the linear-elastic range. [2]

ε_{rod} Rod strain, dimensionless.
E Steel Young's modulus, Pa.
σ_{rod} Axial tensile stress, Pa.

$$\varepsilon_{rod} = \frac{\sigma_{rod}}{E} \tag{10.4}$$

$$\varepsilon_{rod} = \frac{1 \cdot 10^{+8}}{2{,}1 \cdot 10^{+11}} = 0{,}00047619 \tag{10.4a}$$

Rod elongation is determined from its force, length and axial stiffness EA. [2]

ΔL_{rod} Rod extension, mm.
A_{rod} Rod cross-sectional area, m².
E Steel Young's modulus, Pa.
F_{rod} Rod axial force, N.
L_{rod} Rod length, m.

$$\Delta L_{rod} = \frac{F_{rod} \cdot L_{rod}}{(E \cdot A_{rod})} \tag{10.5}$$

$$\Delta L_{rod} = \frac{10000 \cdot 0{,}5}{\left(2{,}1 \cdot 10^{+11} \cdot 1 \cdot 10^{-4}\right)} = 0{,}0002381 \text{ m} = 0{,}2381 \text{ mm} \tag{10.5a}$$

Spring index is mean coil diameter divided by wire diameter; this model requires an index of at least 4. [41]

c_{sp} Spring index, dimensionless.
D_{sp} Mean spring diameter, m.
d_w Spring wire diameter, m.

$$c_{sp} = \frac{D_{sp}}{d_w} \tag{10.6}$$

$$c_{sp} = \frac{0{,}03}{0{,}004} = 7{,}5 \tag{10.6a}$$

The wire curvature factor uses Chassis Handbook equation 3.57; it corrects stress relative to a straight rod. [41]

$k_{\tau,sp}$ Wire curvature factor, dimensionless.
c_{sp} Spring index, dimensionless.

$$k_{\tau,sp} = \frac{(c_{sp} + 0.5)}{(c_{sp} - 0.75)} \tag{10.7}$$

$$k_{\tau,sp} = \frac{(7{,}5 + 0{,}5)}{(7{,}5 - 0{,}75)} = 1{,}1852 \tag{10.7a}$$

The cylindrical spring rate uses round wire and the specified active coil count. [41]

k_{sp} Spring rate, N/mm.
D_{sp} Mean spring diameter, m.
G Shear modulus, Pa.
N_{sp} Active coils, dimensionless.
d_w Spring wire diameter, m.

$$k_{sp} = \frac{G \cdot {d_w}^4}{\left(8 \cdot {D_{sp}}^3 \cdot N_{sp}\right)} \tag{10.8}$$

$$k_{sp} = \frac{8{,}07692 \cdot 10^{+10} \cdot 0{,}004^4}{\left(8 \cdot 0{,}03^3 \cdot 8\right)} = 11966 \text{ N/m} = 11{,}966 \text{ N/mm} \tag{10.8a}$$

Working deflection follows from linear stiffness at the specified working force. [41]

f_{sp} Spring working deflection, mm.
F_{sp} Spring force, N.
k_{sp} Spring rate, N/m.

$$f_{sp} = \frac{F_{sp}}{k_{sp}} \tag{10.9}$$

$$f_{sp} = \frac{100}{11965{,}8} = 0{,}0083571 \text{ m} = 8{,}3571 \text{ mm} \tag{10.9a}$$

Wire shear stress includes curvature correction. Check coil contact and fatigue life separately. [41]

τ_{sp} Wire shear stress, MPa.

D_{sp} Mean spring diameter, m.
F_{sp} Spring force, N.
$k_{\tau,sp}$ Wire curvature factor, dimensionless.
d_w Spring wire diameter, m.
π number π, dimensionless.

$$\tau_{sp} = \frac{k_{\tau,sp} \cdot 8 \cdot F_{sp} \cdot D_{sp}}{\left(\pi \cdot {d_w}^3\right)} \tag{10.10}$$

$$\tau_{sp} = \frac{1{,}18519 \cdot 8 \cdot 100 \cdot 0{,}03}{\left(3{,}14159 \cdot 0{,}004^3\right)} = 141470000\ \text{Pa} = 141{,}47\ \text{MPa} \tag{10.10a}$$

Use the entered allowable shear stress or calculate it from the material limit and required safety factor. [40]

$[\tau]_{sp}$ Allowable wire shear stress, MPa.
R_{sp} Wire-material limiting shear stress, Pa.
n_{sp} Required spring safety factor [n], dimensionless.

$$[\tau]_{sp} = \frac{R_{sp}}{n_{sp}} \tag{10.11}$$

$$[\tau]_{sp} = \frac{6 \cdot 10^{+8}}{1{,}5} = 400000000\ \text{Pa} = 400\ \text{MPa} \tag{10.11a}$$

Compare the actual safety factor for the selected shear criterion with required [n] > 1; this check does not replace fatigue assessment. [40]

n_{sp} Actual spring safety factor, dimensionless.
R_{sp} Wire-material limiting shear stress, Pa.
τ_{sp} Wire shear stress, Pa.

$$n_{sp} = \frac{R_{sp}}{\tau_{sp}} \tag{10.12}$$

$$n_{sp} = \frac{6 \cdot 10^{+8}}{1{,}41471 \cdot 10^{+8}} = 4{,}2412 \tag{10.12a}$$

Strain energy is the area under the linear spring force–deflection curve. [41]

W_{sp} Spring strain energy, J.
F_{sp} Spring force, N.
f_{sp} Spring working deflection, m.

$$W_{sp} = \frac{F_{sp} \cdot f_{sp}}{2} \tag{10.13}$$

$$W_{sp} = \frac{100 \cdot 0{,}00835714}{2} = 0{,}41786 \text{ J} \tag{10.13a}$$

Convert spring force to wheel force at a constant travel ratio using virtual work equality. [41]

F_w Wheel force at constant motion ratio, N.
F_{sp} Spring force, N.
i_{motion} Spring travel / wheel travel ratio, dimensionless.

$$F_w = F_{sp} \cdot i_{motion} \tag{10.14}$$

$$F_w = 100 \cdot 0{,}8 = 80 \text{ N} \tag{10.14a}$$

Spring rate converts to wheel rate using the square of the constant motion ratio. [41]

k_w Wheel rate, N/mm.
i_{motion} Spring travel / wheel travel ratio, dimensionless.
k_{sp} Spring rate, N/m.

$$k_w = k_{sp} \cdot {i_{motion}}^2 \tag{10.15}$$

$$k_w = 11965{,}8 \cdot 0{,}8^2 = 7658{,}1 \text{ N/m} = 7{,}6581 \text{ N/mm} \tag{10.15a}$$

11. Comparison and design justification

Save a baseline with the same working-machine requirements. “Prepare construction refinement” shows the initial failed condition, adopted change and repeated calculations. Acceptance updates the fields and saves the sequence in JSON and PDF. If no catalogue solution exists, the condition remains failed. State the selection reason, limit comparison, actual safety factor and model limitations in the conclusion.

Appendix. Verified teaching examples

With the same requirements $T = 200\ \text{N·m}$, $n = 76{,}394\ \text{min}^{-1}$ and a 2.2 kW / 1440 min^{-1} motor, the flexible transmission changes gearbox input speed. Automatic selection refines the allocation between both stages and the flexible transmission; all other changes require a repeated check.

Table A.1 — Results after adopting integer tooth counts

Assembly	**z_3**	**Speed, min^{-1}**	**Deviation / tolerance**	**Torque, N·m**
No flexible transmission	69	75,889	δ_n = -0,6611 % ∈ [−5 %; +5 %]	263,21
Belt drive	48	76,923	δ_n = 0,69207 % ∈ [−5 %; +5 %]	246,69
Timing belt	42	76,759	δ_n = 0,47738 % ∈ [−5 %; +5 %]	252,42
Chain drive	45	76,768	δ_n = 0,48865 % ∈ [−5 %; +5 %]	249,79

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